Solved selected problems of Introductory functional analysis with applications - Erwin Kreyszig

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Chapter 3 - Inner Product Spaces. Hilbert Spaces

3.1 - Inner Product Space. Hilbert Space

Proof. **1** Let us prove (4) as follows

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y\rangle + \langle x-y, x-y\rangle \\
&= \langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle + \langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle \\
&= 2\langle x,x\rangle + 2\langle y,y\rangle \\
&= 2(\|x\|^2 + \|y\|^2)
\end{aligned}$$

Where we used the definition of the norm $\|x\| = \sqrt{\langle x,x\rangle}$ and the distributive property of the inner products. $\square$

Proof. **2** Let $x \perp y$ in an inner product space X then

$$\begin{aligned}
\|x+y\|^2 &= \langle x+y, x+y\rangle \\
&= \langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle \\
&= \langle x,x\rangle + \langle y,y\rangle \\
&= \|x\|^2 + \|y\|^2
\end{aligned}$$

Where we used that $\langle x,y\rangle = \langle y,x\rangle = 0$ since $x \perp y$.

Suppose now we have m mutually orthogonal vectors, i.e. $x_1 \perp x_2 \perp ... \perp x_m$ then

$$\begin{aligned}
\|x_1 + x_2 + ... + x_m\|^2 &= \langle x_1 + x_2 + ... + x_m, x_1 + x_2 + ... + x_m\rangle \\
&= \langle x_1, x_1\rangle + \langle x_1, x_2\rangle + ... + \langle x_m, x_{m-1}\rangle + \langle x_m, x_m\rangle \\
&= \langle x_1, x_1\rangle + \langle x_2, x_2\rangle + ... + \langle x_m, x_m\rangle \\
&= \|x_1\|^2 + \|x_2\|^2 + ... + \|x_m\|^2
\end{aligned}$$

Where we used that $\langle x_i, x_j\rangle = 0$ for $i \neq j$ since they are orthogonal. $\square$

Proof. **3** Let $x, y \in \mathbb{R}$ and suppose we have the following property

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2$$

then since $\|x\|^2 = \langle x, x\rangle$ above equation implies that

$$\begin{aligned} \langle x+y, x+y\rangle &= \langle x, x\rangle + \langle y, y\rangle \\ \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle &= \langle x, x\rangle + \langle y, y\rangle \\ \langle x, y\rangle + \langle y, x\rangle &= 0 \end{aligned}$$

But since $x, y \in \mathbb{R}$ we have that $\langle x, y\rangle = \langle y, x\rangle$, hence

$$\langle x, y\rangle = \langle y, x\rangle = 0$$

Therefore $x \perp y$.

Let now $x, y \in \mathbb{C}$ and suppose that the property holds in this case too. Then as before must be that

$$\langle x, y\rangle + \langle y, x\rangle = 0$$

Also, we know that $\langle y, x\rangle = \overline{\langle x, y\rangle}$ hence we need that

$$\langle x, y\rangle = -\overline{\langle x, y\rangle}$$

But since we are assuming a complex inner product we have that $\langle x, y\rangle$ is of the form $a + bi$ hence

$$a + bi = -(a - bi) = -a + bi$$

Equating the real and imaginary parts we see that $a = 0$ and $b = b$. So for example if $x = 1 + i$ and $y = 1 - i$ we see that the Pythagorean property holds, as we show below

$$\|1 + i + 1 - i\|^2 = \|2\|^2 = 4 = 2 + 2 = \|1 + i\|^2 + \|1 - i\|^2$$

But inner product gives us

$$\langle x, y\rangle = (1 + i)\overline{(1 - i)} = (1 + i)(1 + i) = 1 + i + i - 1 = 2i$$

Therefore x and y are not orthogonal. $\square$

Proof. **6** Let $x \neq 0$ and $y \neq 0$

(a) Let $x \perp y$ we want to prove that $\{x, y\}$ is a linearly independent set i.e. we want to prove

$$\alpha_1 x + \alpha_2 y = 0$$

Only for $\alpha_1 = \alpha_2 = 0$. Suppose the opposite we want to arrive at a contradiction, let $\alpha_1 \neq 0$ and $\alpha_2 = 0$ then $\alpha_1 x + 0y = 0$ i.e. $x = 0$ but we said that $x \neq 0$ so it must be that $\alpha_1 \neq 0$ and $\alpha_2 \neq 0$. Then we can write that

$$x = -\frac{\alpha_2}{\alpha_1} y$$

Also, we know that $\langle x, y \rangle = 0$ since $x \perp y$ then

$$\langle x, y \rangle = \left\langle -\frac{\alpha_2}{\alpha_1} y, y \right\rangle = -\frac{\alpha_2}{\alpha_1} \langle y, y \rangle = 0$$

But this implies that $y = 0$ and we said that $y \neq 0$, a contradiction.

Therefore must be that $\{x, y\}$ is a linearly independent set.

(b) Now, let $\{x_1, ..., x_m\}$ be a set of mutually orthogonal non-zero vectors, we want to show they are a linearly independent set.

As before, suppose the opposite, we want to arrive at a contradiction.

If $\alpha_i \neq 0$ for only the ith vector then $x_i = 0$ and that cannot happen.

So let us assume that $\alpha_i \neq 0$ for $1 \leq i \leq m$, if not all α_is were different from zero the following still applies, then

$$\begin{aligned} \alpha_1 x_1 + ... + \alpha_n x_n = 0 \\ x_1 = -\frac{\alpha_2}{\alpha_1} x_2 - ... - \frac{\alpha_n}{\alpha_1} x_n \end{aligned}$$

Also we know that $\langle x_1, x_2 \rangle = 0$ then

$$\begin{aligned} \langle x_1, x_2 \rangle &= \left\langle -\frac{\alpha_2}{\alpha_1} x_2 - ... - \frac{\alpha_n}{\alpha_1} x_n, x_2 \right\rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle - ... - \frac{\alpha_n}{\alpha_1} \langle x_n, x_2 \rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle \\ &= 0 \end{aligned}$$

Where we used that $\langle x_i, x_2 \rangle = 0$ for $1 \leq i \leq m$ and $i \neq 2$. But the last equality implies that $x_2 = 0$ which is a contradiction.

Therefore must be that the orthogonal nonzero vectors $\{x_1, ..., x_m\}$ are a linearly independent set.

$\square$

Proof. **7** Let $\langle x, u\rangle = \langle x, v\rangle$ for all x, then must also be true for $x = u$ i.e. $\langle u, u\rangle = \langle u, v\rangle$ and for $x = v$ i.e. $\langle v, u\rangle = \langle v, v\rangle$.
Now, let us compute $\langle u - v, u - v\rangle$ as follows

$$\begin{aligned}\langle u - v, u - v\rangle &= \langle u, u\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, v\rangle \\ &= \langle u, v\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, u\rangle \\ &= 0\end{aligned}$$

But then this implies that $u - v = 0$. Therefore $u = v$. □

Proof. **8** Let us compute (9) knowing that $\|x\| = \sqrt{\langle x, x\rangle}$ then

$$\begin{aligned}\frac{1}{4}(\|x + y\|^2 - \|x - y\|^2) &= \frac{1}{4}(\langle x + y, x + y\rangle - \langle x - y, x - y\rangle) \\ &= \frac{1}{4}(\langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle - \\ &\quad - (\langle x, x\rangle + \langle x, -y\rangle + \langle -y, x\rangle + \langle -y, -y\rangle)) \\ &= \frac{1}{4}(\langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle - \\ &\quad - \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle - \langle y, y\rangle) \\ &= \frac{1}{4}(2\langle x, y\rangle + 2\langle x, y\rangle) \\ &= \langle x, y\rangle\end{aligned}$$

Where we used that $\langle x, y\rangle = \langle y, x\rangle$ since we are in a real inner product space. □

Proof. **9** Let us compute (10) knowing that we are in a complex inner product space, then

$$\begin{aligned}
\frac{1}{4}(\|x+y\|^2 - \|x-y\|^2) &= \frac{1}{4}(\langle x+y, x+y\rangle - \langle x-y, x-y\rangle) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\overline{\langle x,y\rangle}) \\
&= \frac{1}{2}(\mathrm{Re}\langle x,y\rangle + i\mathrm{Im}\langle x,y\rangle + \mathrm{Re}\langle x,y\rangle - i\mathrm{Im}\langle x,y\rangle) \\
&= \mathrm{Re}\langle x,y\rangle
\end{aligned}$$

Now, we compute the second equation as follows

$$\begin{aligned}
\frac{1}{4}(\|x+iy\|^2 - \|x-iy\|^2) &= \frac{1}{4}(\langle x+iy, x+iy\rangle - \langle x-iy, x-iy\rangle) \\
&= \frac{1}{4}(\langle x,x\rangle + \langle x,iy\rangle + \langle iy,x\rangle + \langle iy,iy\rangle - \\
&\quad - (\langle x,x\rangle + \langle x,-iy\rangle + \langle -iy,x\rangle + \langle -iy,-iy\rangle)) \\
&= \frac{1}{4}(\langle x,x\rangle - i\langle x,y\rangle + i\langle y,x\rangle + \langle y,y\rangle - \\
&\quad - \langle x,x\rangle - i\langle x,y\rangle + i\langle y,x\rangle - \langle y,y\rangle) \\
&= \frac{1}{2}(-i\langle x,y\rangle + i\langle y,x\rangle) \\
&= \frac{1}{2}(-i\langle x,y\rangle + i\overline{\langle x,y\rangle}) \\
&= \frac{1}{2}(-i\mathrm{Re}\langle x,y\rangle + \mathrm{Im}\langle x,y\rangle + i\mathrm{Re}\langle x,y\rangle + \mathrm{Im}\langle x,y\rangle) \\
&= \mathrm{Im}\langle x,y\rangle
\end{aligned}$$

$\square$

Proof. **10** Let z_1 and z_2 be complex numbers and let $\langle z_1, z_2\rangle = z_1\overline{z_2}$ we want to prove that in fact this is a valid definition of an inner product.
So if this expression defines an inner product, it must have the four properties of inner products. Then for (IP1) let z_3 be also a complex number

$$\langle z_1 + z_3, z_2\rangle = (z_1 + z_3)\overline{z_2} = z_1\overline{z_2} + z_3\overline{z_2} = \langle z_1, z_2\rangle + \langle z_3, z_2\rangle$$

For (IP2) we see that

$$\langle \alpha z_1, z_2\rangle = \alpha z_1\overline{z_2} = \alpha \langle z_1, z_2\rangle$$

For (IP3) we see that

$$\langle z_1, z_2\rangle = z_1\overline{z_2} = \overline{\overline{z_1}z_2} = \overline{\langle z_2, z_1\rangle}$$

Finally for (IP4) we see that $\langle z_1, z_1\rangle = z_1\overline{z_1} \geq 0$. Also, let $\langle z_1, z_1\rangle = 0$ then

$$z_1\overline{z_1} = (x + iy)(x - iy) = x^2 + y^2 = 0$$

Then the only way for this equation to hold is that $x = y = 0$ and hence $z_1 = 0$. Now, if we suppose $z_1 = 0$ then $\langle z_1, z_1\rangle = z_1\overline{z_1} = 0$ and hence (IP4) is also satisfied.
Therefore $\langle z_1, z_1\rangle = z_1\overline{z_1}$ defines in fact an inner product.
Also, to have orthogonality it must happen that $\langle z_1, z_2\rangle = z_1\overline{z_2} = 0$. $\square$

Proof. **11** Let X be the vector space of all ordered pairs of complex numbers. The norm defined as $\|x\| = |\xi_1| + |\xi_2|$ cannot be obtained from an inner product. We prove this by showing that the norm does not satisfy the parallelogram equality. Let us take $x = (-1, 1)$ and $y = (-1, -1)$ then

$$2(\|x\|^2 + \|y\|^2) = 2((|-1| + |1|)^2 + (|-1| + |-1|)^2) = 16$$

But

$$\|x + y\|^2 + \|x - y\|^2 = (|-1-1| + |1-1|)^2 + (|-1+1| + |1+1|)^2 = 8$$

Therefore the parallelogram equality is not satisfied and hence the norm defined cannot be obtained from an inner product. □

Proof. **13** The inner product defined in 3.1-5 for continuous functions states that

$$\langle x, y \rangle = \int_a^b x(t)\overline{y(t)}\ dt$$

Let us check that this definition satisfies (IP1) to (IP4). We assume $x(t)$ and $y(t)$ complex-valued functions.
(IP1):

$$\begin{aligned}
\langle x + y, z \rangle &= \int_a^b (x(t) + y(t))\overline{z(t)}\ dt \\
&= \int_a^b x(t)\overline{z(t)}\ dt + \int_a^b y(t)\overline{z(t)}\ dt \\
&= \langle x, z \rangle + \langle y, z \rangle
\end{aligned}$$

(IP2):

$$\langle \alpha x, y \rangle = \int_a^b \alpha x(t)\overline{y(t)}\ dt = \alpha \int_a^b x(t)\overline{y(t)}\ dt = \alpha \langle x, y \rangle$$

(IP3):

$$\langle x, y \rangle = \int_a^b x(t)\overline{y(t)}\ dt = \int_a^b \overline{\overline{x(t)}y(t)}\ dt = \overline{\langle y, x \rangle}$$

(IP4):

$$\langle x, x \rangle = \int_a^b x(t)\overline{x(t)}\ dt = \int_a^b |x(t)|^2\ dt \geq 0$$

Let now $\langle x, x \rangle = 0$ then

$$\int_a^b |x(t)|^2\ dt = 0$$

Then must be that $x(t) = 0$ since $x(t)$ is continuous. Conversely, if we suppose that $x(t) = 0$ we get that

$$\langle x, x \rangle = \int_a^b 0 \cdot 0\ dt = 0$$

Therefore the inner product defined in 3.1-5 satisfy the properties (IP1) to (IP4). $\square$

3.2 - Further Properties of Inner Produet Spaces

Proof. **2** An example of subspace of l^2 is the set sequences with finitely many entries different from zero followed by zeroes i.e.

$$Y = \{(x_1, x_2, ..., x_k, 0, 0, 0, ...) : x_i \in \mathbb{R}, 0 \leq i \leq k\}$$

Suppose $x = (x_1, x_2, ..., x_k, 0, 0, 0, ...) \in Y$ then

$$\|x\|_2 = \left(\sum_{i=1}^{\infty} |x_i|^2\right)^{1/2} = \left(\sum_{i=1}^{k} |x_i|^2\right)^{1/2} < \infty$$

So $Y \subset l^2$.
On the other hand, let $x' = (x'_1, x'_2, ..., x'_k, 0, 0, 0, ...) \in Y$, and α, β be scalars then

$$\begin{aligned}\alpha x + \beta x' &= \alpha(x_1, x_2, ..., x_k, 0, 0, 0, ...) + \beta(x'_1, x'_2, ..., x'_k, 0, 0, 0, ...)\\ &= (\alpha x_1, \alpha x_2, ..., \alpha x_k, 0, 0, 0, ...) + (\beta x'_1, \beta x'_2, ..., \beta x'_k, 0, 0, 0, ...)\\ &= (\alpha x_1 + \beta x'_1, \alpha x_2 + \beta x'_2, ..., \alpha x_k + \beta x'_k, 0, 0, 0, ...)\end{aligned}$$

We see that $\alpha x_i + \beta x'_i \in \mathbb{R}$ since $\mathbb{R}$ is a vector space then Y is a subspace of l^2. $\square$

Proof. **3** Let X be the inner product space consisting of the polynomial $x = 0$ and all real polynomials in t of degree not exceeding 2, considered for real $t \in [a, b]$ with inner product defined as

$$\langle x, y \rangle = \int_a^b x(t)y(t)\ dt$$

Let $(x_n) = (\alpha_n t^2 + \beta_n t + \gamma_n) \in X$ be a Cauchy sequence on X where $\alpha_n, \beta_n, \gamma_n \in \mathbb{R}$ then since this is a Cauchy sequence, given $\epsilon > 0$ we can find $N \in \mathbb{N}$ such that when $n, m > N$ we have that

$$\|x_n - x_m\| = \left(\int_a^b (x_n(t) - x_m(t))^2\ dt \right)^{1/2} < \epsilon$$

Hence

$$\left(\int_a^b ((\alpha_n - \alpha_m)t^2 + (\beta_n - \beta_m)t + (\gamma_n - \gamma_m))^2\ dt \right)^{1/2} < \epsilon$$

For this integral to tend to 0 over the fixed interval $[a, b]$ since t does not depend on n must happen that $(\alpha_n - \alpha_m) \to 0$, $(\beta_n - \beta_m) \to 0$ and $(\gamma_n - \gamma_m) \to 0$.
This implies that (α_n), (β_n) and (γ_n) are Cauchy sequences of real numbers which we know they converge, let us say that they converge to α, β and γ respectively.
Now, let us define $x = \alpha t^2 + \beta t + \gamma$ then

$$\begin{aligned} \|x_n - x\| &= \left(\int_a^b (x_n(t) - x(t))^2\ dt \right)^{1/2} \\ &= \left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2\ dt \right)^{1/2} \end{aligned}$$

But we know that $\alpha_n \to \alpha$, $\beta_n \to \beta$ and $\gamma_n \to \gamma$ therefore we have that

$$\left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2\ dt \right)^{1/2} \to 0$$

Or what is the same $x_n \to x$ and hence since x_n was a Cauchy sequence then X is complete.

Let now Y be all $x \in X$ such that $x(a) = 0$. Let $y, y' \in Y$ and let α, β be scalars. We know that when $t = a$ then $y(a) = y'(a) = 0$ then for $z(t) = \alpha y(t) + \beta y'(t)$ we have that $z(a) = \alpha y(a) + \beta y'(a) = 0 + 0 = 0$. Therefore Y is a subspace of X.

Finally, let Z be all $x \in X$ of degree 2. Let $z, z' \in Z$ and let λ, ρ be scalars. Then let us compute $\lambda z + \rho z'$ as follows

$$\lambda(\alpha t^2 + \beta t + \gamma) + \rho(\alpha' t^2 + \beta' t + \gamma') = (\lambda\alpha + \rho\alpha')t^2 + (\lambda\beta + \rho\beta')t + (\lambda\gamma + \rho\gamma')$$

Where $\alpha, \alpha', \beta, \beta', \gamma$ and γ' are not zero. Then we see that if $\lambda = \rho = 0$ we get that $\lambda z + \rho z' = 0$ but the polynomial $z = 0$ is not included in Z then Z is not a subspace of X. □

Proof. **4** Let $y \perp x_n$ and $x_n \to x$ then by the Continuity of the Inner Product we have that

$$\langle x_n, y \rangle \to \langle x, y \rangle$$

This implies that

$$\| \langle x_n, y \rangle - \langle x, y \rangle \| \to 0$$

But we know that $\langle x_n, y \rangle = 0$ because $y \perp x_n$ hence

$$\| \langle x, y \rangle \| \to 0$$

And by the (N2) property for norms, we get that

$$\langle x, y \rangle = 0$$

Therefore $x \perp y$. □

Proof. **5** Let a sequence (x_n) in an inner product space such that $\|x_n\| \to \|x\|$ and $\langle x_n, x\rangle \to \langle x, x\rangle$. Given that $\|x_n\| \to \|x\|$ then $\|x_n\|^2 \to \|x\|^2$ so given $\epsilon^2/3 > 0$ there is $N \in \mathbb{N}$ such that when $n \geq N$ we see that

$$|\|x_n\|^2 - \|x\|^2| = |\langle x_n, x_n\rangle - \langle x, x\rangle| < \epsilon^2/3$$

On the other hand, since $\langle x_n, x\rangle \to \langle x, x\rangle$ given $\epsilon^2/3 > 0$ there is $N' \in \mathbb{N}$ such that when $n \geq N'$ we get that

$$|\langle x_n, x\rangle - \langle x, x\rangle| = |\langle x_n - x, x\rangle| < \epsilon^2/3$$

Also, let us note that

$$\begin{aligned}
\|x_n - x\|^2 &= |\langle x_n - x, x_n - x\rangle| \\
&= |\langle x_n - x, x_n\rangle - \langle x_n - x, x\rangle| \\
&\leq |\langle x_n - x, x_n\rangle| + |\langle x_n - x, x\rangle| \\
&\leq |\langle x_n, x_n\rangle - \langle x, x_n\rangle| + |\langle x_n - x, x\rangle| \\
&\leq |\langle x_n, x_n\rangle - \langle x, x\rangle + \langle x, x\rangle - \langle x, x_n\rangle| + |\langle x_n - x, x\rangle| \\
&\leq |\langle x_n, x_n\rangle - \langle x, x\rangle| + |\langle x, x - x_n\rangle| + |\langle x_n - x, x\rangle|
\end{aligned}$$

Then if we take $M = \max(N, N')$ when $n \geq M$ we have that

$$\begin{aligned}
\|x_n - x\|^2 &\leq |\langle x_n, x_n\rangle - \langle x, x\rangle| + |\langle x, x - x_n\rangle| + |\langle x_n - x, x\rangle| \\
&< \epsilon^2/3 + \epsilon^2/3 + \epsilon^2/3 = \epsilon^2
\end{aligned}$$

Therefore $\|x_n - x\| < \epsilon$ and hence $x_n \to x$. $\square$

Proof. **9** Let V be the vector space of all continuous complex-valued functions on $J = [a, b]$. Let $X_1 = (V, \|\cdot\|_\infty)$ where $\|x\|_\infty = \max_{t\in J} |x(t)|$ and let $X_2 = (V, \|\cdot\|_2)$ where

$$\|x\|_2 = \sqrt{\langle x, x\rangle} \qquad \langle x, y\rangle = \int_a^b x(t)\overline{y(t)}\; dt$$

Also, let us define $i : X_1 \to X_2$ be the identity mapping $x \to x$.
Suppose we take $x, y \in V$ then we see that

$$\begin{aligned}\|x - y\|_2 &= \sqrt{\langle x - y, x - y\rangle} \\ &= \sqrt{\int_a^b (x(t) - y(t))(\overline{x(t) - y(t)})\; dt} \\ &= \sqrt{\int_a^b |x(t) - y(t)|^2\; dt}\end{aligned}$$

But also we have that

$$\sqrt{\int_a^b |x(t) - y(t)|^2\; dt} \leq \sqrt{\int_a^b \max_{t\in J} |x(t) - y(t)|^2\; dt} \;\; = \max_{t\in J} |x(t) - y(t)|\sqrt{b - a}$$

So given $\epsilon > 0$ if we take $\delta = \epsilon/\sqrt{b - a}$ such that

$$\|x - y\|_\infty = \max_{t\in J} |x(t) - y(t)| < \delta = \frac{\epsilon}{\sqrt{b - a}}$$

We get that

$$\|x - y\|_2 \leq \max_{t\in J} |x(t) - y(t)|\sqrt{b - a} < \epsilon$$

Therefore the identity mapping i is continuous.

$\square$

Proof. **10** Let $T : X \to X$ be a bounded linear operator on a complex inner product space X and let $\langle Tx, x\rangle = 0$ for all $x \in X$.
Let $x = \alpha u + v \in X$ for some $u, v \in X$ and $\alpha \in \mathbb{C}$ then

$$\begin{aligned}\langle Tx, x\rangle &= \langle \alpha Tu + Tv, \alpha u + v\rangle \\ &= \langle \alpha Tu, \alpha u + v\rangle + \langle Tv, \alpha u + v\rangle \\ &= \alpha\overline{\alpha} \langle Tu, u\rangle + \alpha \langle Tu, v\rangle + \overline{\alpha} \langle Tv, u\rangle + \langle Tv, v\rangle \\ &= \alpha \langle Tu, v\rangle + \overline{\alpha} \langle Tv, u\rangle\end{aligned}$$

Then taking $\alpha = 1$ we get that

$$\begin{aligned}\langle Tu, v\rangle + \langle Tv, u\rangle &= 0 \\ \langle Tu, v\rangle &= -\langle Tv, u\rangle\end{aligned}$$

But if we take $\alpha = i$ we have that $i \langle Tu, v\rangle - i \langle Tv, u\rangle = 0$ or

$$\begin{aligned}\langle Tu, v\rangle - \langle Tv, u\rangle &= 0 \\ \langle Tu, v\rangle &= \langle Tv, u\rangle\end{aligned}$$

This implies that $\langle Tu, v\rangle = 0$ for all $u, v \in X$ so in particular if we select $v = Tu$ then $\langle Tu, Tu\rangle = 0$ and hence by (IP4) we have that $Tu = 0$. Therefore $T = 0$.

Let us consider now the Euclidean plane, a real inner product space, and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos \pi/2 & -\sin \pi/2 \\ \sin \pi/2 & \cos \pi/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

Then let us compute the following inner product

$$\left\langle T \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} -y \\ x \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = -yx + xy = 0$$

Therefore, we see that the inner product is 0 but $T \neq 0$ so above result does not hold for real inner product spaces. $\square$

3.3 - Orthogonal Complements and Direct Sums

Proof. **1** Let H be a Hilbert space, $M \subset H$ a convex subset, and (x_n) a sequence in M such that $\|x_n\| \to d$ where $d = \inf_{x \in M} \|x\|$.
Let us consider $\overline{M}$ the closure of M then $\overline{M}$ is complete because $\overline{M} \subset H$, also, $(x_n) \in \overline{M}$.
Let $n, m \in \mathbb{N}$ where $n \geq m$. We see that $\frac{1}{2}x_n + \frac{1}{2}x_m \in M$ because M is convex but also we have that

$$\left\| \frac{x_n + x_m}{2} \right\| \geq d = \inf_{x \in M} \|x\|$$

Hence

$$\|x_n + x_m\| \geq 2d$$

On the other hand, by the parallelogram equality we get that

$$\begin{aligned} \|x_n - x_m\|^2 &= -\|x_n + x_m\|^2 + 2(\|x_n\|^2 + \|x_m\|^2) \\ &\leq -(2d)^2 + 2(\|x_n\|^2 + \|x_m\|^2) \end{aligned}$$

But since $\|x_n\| \to d$ as $n, m \to \infty$ then $\|x_n - x_m\| \to 0$. Therefore (x_n) is Cauchy, but this also implies that (x_n) converges because $\overline{M}$ is a subset of a Hilbert space. $\square$

Proof. **5** Let $X = \mathbb{R}^2$ then

(a) Let $M = \{x\}$ where $x = (\xi_1, \xi_2) \neq 0$ then $M^\perp$ by definition is

$$M^\perp = \{x' \in X \mid x' \perp M\} = \{x' \in X \mid \langle x', x \rangle = 0\}$$

Let $x' = (-\xi_2, \xi_1) \in X$ then we see that

$$\langle x', x \rangle = -\xi_1\xi_2 + \xi_2\xi_1 = 0$$

So $(-\xi_2, \xi_1) \in M^\perp$ but $(-\alpha\xi_2, \alpha\xi_1)$ for $\alpha \in \mathbb{R}$ is also in $M^\perp$. Therefore $M^\perp = \{(-\alpha\xi_2, \alpha\xi_1) \in X \mid \alpha \in \mathbb{R}\}$.

(b) Let $M = \{x_1, x_2\}$ where x_1 and x_2 are linearly independent then

$$\begin{aligned} M^\perp &= \{x' \in X \mid x' \perp M\} \\ &= \{x' \in X \mid \langle x', x_1 \rangle = 0 \text{ and } \langle x', x_2 \rangle = 0\} \end{aligned}$$

Suppose $x' \in X$ satisfy $\langle x', x_1 \rangle = 0$ and $\langle x', x_2 \rangle = 0$, then we have that

$$\langle x', x_1 \rangle = \eta_1\xi_{11} + \eta_2\xi_{12} = 0$$

and

$$\langle x', x_2 \rangle = \eta_1\xi_{21} + \eta_2\xi_{22} = 0$$

Where we assumed that $x_1 = (\xi_{11}, \xi_{12})$, $x_2 = (\xi_{21}, \xi_{22})$ and $x' = (\eta_1, \eta_2)$. But since x_1 and x_2 are linearly independent then the equations

$$\begin{aligned} c_1\xi_{11} + c_2\xi_{21} &= 0 \\ c_1\xi_{12} + c_2\xi_{22} &= 0 \end{aligned}$$

have only one solution, $c_1 = c_2 = 0$.
Therefore must be that $x' = 0$ and hence

$$M^\perp = \{0\}$$

$\square$

Proof. **6**
Let $Y = \{x \mid x = (\xi_j) \in l^2, \xi_{2n} = 0, n \in \mathbb{N}\}$ we want to show that Y is a closed subspace of l^2.
Let $(x_j) \subset Y$ and suppose it converges to some $x = (\eta_j) \notin Y$ we want to arrive at a contradiction, then given $\epsilon > 0$ there is some $N \in \mathbb{N}$ such that when $n \geq N$ we have that $\|x_n - x\| < \epsilon$, also, we see that

$$\|x_n - x\| = \sqrt{\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2} = \sqrt{\sum_{j=1}^{\infty} |\xi_{2j+1} - \eta_{2j+1}|^2 + \sum_{j=1}^{\infty} |\eta_{2j}|^2}$$

Since the first term on the RHS is non-negative then

$$\|x_n - x\| \geq \sqrt{\sum_{j=1}^{\infty} |\eta_{2j}|^2}$$

So we have a lower bound on $\|x_n - x\|$ but $\|x_n - x\| \to 0$, a contradiction. Therefore, must be that $\eta_{2j} = 0$ so $x \in Y$ and hence Y is closed.

Given $x \in Y$, to find $Y^\perp$ we need to find all elements $z \in l^2$ such that $\langle x, z\rangle = 0$. We see that

$$\langle x, z\rangle = \sum_{j=1}^{\infty} \xi_j \bar{\eta}_j = \sum_{j=1}^{\infty} \xi_{2j+1}\bar{\eta}_{2j+1} + \sum_{j=1}^{\infty} \xi_{2j}\bar{\eta}_{2j} = \sum_{j=1}^{\infty} \xi_{2j+1}\bar{\eta}_{2j+1}$$

So for $\langle x, z\rangle = 0$ then must be that $\bar{\eta}_{2j+1} = 0$ because it must hold for every $x \in Y$. Therefore

$$Y^\perp = \{x \mid x = (\eta_j) \in l^2, \eta_{2n+1} = 0, n \in \mathbb{N}\}$$

Finally, let $Y = \text{span}\{e_1, ..., e_n\} \subset l^2$ where $e_j = (\delta_{jk})$.
We see that an element $x \in Y$ can be written as

$$x = \alpha_1 e_1 + ... + \alpha_n e_n = \Big(\alpha_1, \alpha_2, ..., \alpha_n, 0, 0, ...\Big)$$

Where $\alpha_1, ..., \alpha_n \in \mathbb{C}$.
Let then $z \in l^2$, we see that

$$\langle x, z\rangle = \alpha_1\bar{\eta}_1 + \alpha_2\bar{\eta}_2 + ... + \alpha_n\bar{\eta}_n + 0 \cdot \bar{\eta}_{n+1} + 0 \cdot \bar{\eta}_{n+2} + ...$$

So for $\langle x, z\rangle = 0$ must be that $\eta_j = 0$ for $1 \leq j \leq n$ and hence

$$Y^\perp = \{x \mid x = (\xi_j) \in l^2 \text{ such that } \xi_j = 0 \text{ for } 1 \leq j \leq n\}$$

$\square$

Proof. **7**

Let A and $A \subset B$ be nonempty subsets of an inner product space X

(a) Let $a \in A$ and $a' \in A^{\perp}$ we see that $\langle a, a' \rangle = 0$ hence $a \perp A^{\perp}$, also by definition we know that

$$A^{\perp\perp} = \{x \in X \mid x \perp A^{\perp}\}$$

Then $a \in A^{\perp\perp}$ and since a was arbitrary we have that

$$A \subset A^{\perp\perp}$$

(b) Let $b \in B$, $a \in A$ and $b' \in B^{\perp}$, by definition $b' \perp B$ hence $\langle b, b' \rangle = 0$ but since a is also in B then must be that $\langle a, b' \rangle = 0$ and since a and b were arbitrary then this implies that $b' \perp A$ and therefore

$$B^{\perp} \subset A^{\perp}$$

(c) From part (a) we have that $A \subset A^{\perp\perp}$ and from part (b) taking $B = A^{\perp\perp}$ we have that

$$B^{\perp} = A^{\perp\perp\perp} \subset A^{\perp}$$

Also, from part (a) if we take A as $A^{\perp}$ following the same steps we get that

$$A^{\perp} \subset A^{\perp\perp\perp}$$

Therefore joining both results we get that $A^{\perp\perp\perp} = A^{\perp}$.

□

Proof. **9**
Let Y be a subspace of a Hilbert space H

($\Rightarrow$) Suppose Y is closed in H then by Lemma 3.3-6 we get that $Y = Y^{\perp\perp}$.

($\Leftarrow$) Suppose $Y = Y^{\perp\perp}$. We prove first that $Y^{\perp}$ is closed in H.
Let $(x_n) \in Y^{\perp}$ be a convergent sequence, that converges to $x \in H$.
Let also, $y \in Y$ then we know that $\langle x_n, y \rangle = 0$, then by continuity of the inner product we have that

$$\langle x_n, y \rangle \to \langle x, y \rangle$$

but $\langle x_n, y \rangle = 0$ then $\langle x, y \rangle = 0$ and hence $x \in Y^{\perp}$. So $Y^{\perp}$ is closed.
We see that $Y^{\perp}$ is a subspace of H since if $y_1', y_2' \in Y^{\perp}$ and $x \in Y$ then

$$\left\langle \alpha y_1' + \beta y_2', x \right\rangle = \alpha \left\langle y_1', x \right\rangle + \beta \left\langle y_2', x \right\rangle = 0$$

So following the same logic we used to prove $Y^{\perp}$ is closed assuming Y is a subspace of H we get that $Y^{\perp\perp}$ is closed but $Y = Y^{\perp\perp}$.
Therefore Y is closed.

$\square$

Proof. **10**
Let $M \neq \emptyset$ be any subset of a Hilbert space H.
Let us notice that $M^{\perp\perp}$ is a closed subspace of H so $M^{\perp\perp}$ is a Hilbert space and hence without loss of generality we may assume that M is a subset of $H = M^{\perp\perp}$.
Let us take $x, y \in H = M^{\perp\perp}$ then $\langle x, y \rangle \neq 0$ because otherwise either x or y are not in $M^{\perp\perp}$ so the only way where $\langle x, y \rangle = 0$ is that $x = y = 0$ and hence $M^{\perp} = \{0\}$.
Then by theorem 3.3-7 we have that the span of M is dense in $M^{\perp\perp}$ but then

$$M \subset \overline{\mathrm{span}(M)} = M^{\perp\perp}$$

Therefore $M^{\perp\perp}$ is the smallest closed subspace containing M. □

3.4 - Orthonormal Sets and Sequences

Proof. **1.** Let X be an inner product space of finite dimension n, then, X contains a lineraly independent set of n vectors (a basis), let us call them $B' = \{b'_1, ..., b'_n\}$.
We want to define $B = \{b_1, ..., b_n\}$ such that the elements are orthonormal. Let us define $b_1 \in X$ such that

$$b_1 = \frac{1}{\|b'_1\|} \cdot b'_1$$

We see that we can write b'_2 as

$$b'_2 = \left\langle b'_2, b_1 \right\rangle b_1 + v_2$$

where v_2 is non-zero since the elements of B' are linearly independent, then

$$v_2 = b'_2 - \left\langle b'_2, b_1 \right\rangle b_1$$

We see that

$$\begin{aligned}
\langle b_1, v_2 \rangle &= \left\langle b_1, b'_2 - \left\langle b'_2, b_1 \right\rangle b_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, \left\langle b'_2, b_1 \right\rangle b_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \overline{\left\langle b'_2, b_1 \right\rangle} \left\langle b_1, b_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, b'_2 \right\rangle \frac{1}{\|b'_1\|^2} \left\langle b'_1, b'_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, b'_2 \right\rangle \frac{\|b'_1\|^2}{\|b'_1\|^2} \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, b'_2 \right\rangle \\
&= 0
\end{aligned}$$

Hence, we see that they are orthogonal. Then we define b_2 as

$$b_2 = \frac{1}{\|v_2\|} \cdot v_2$$

Now, to define b_3 we note that there is v_3 such that

$$v_3 = b'_3 - \left\langle b'_3, b_1 \right\rangle b_1 - \left\langle b'_3, b_2 \right\rangle b_2$$

We see that v_3 is non-zero since the elements of B' are linearly independent. Also, $v_3 \perp b_1$ and $v_3 \perp b_2$, then we define b_3 as

$$b_3 = \frac{1}{\|v_3\|} \cdot v_3$$

We can carry on this process n times until we find

$$v_n = b'_n - \sum_{i=1}^{n-1} \left\langle b'_n, b_i \right\rangle b_i$$

From which we define b_n as

$$b_n = \frac{1}{\|v_n\|} \cdot v_n$$

Therefore we have found a set B that is orthogonal pairwise, where each element has norm equal to 1, so B is an orthonormal set.
Finally, because of Lemma 3.4-2 we know that B is linearly independent so B is a basis for X as well. $\square$

Proof. **2.** Equation (12*) states that

$$\sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Where $x \in \mathbb{R}^r$ and $r \geq n$. We know from equation (9) that

$$\|y\|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2$$

So y is the projection of x onto a set of orthonormal vectors in $\mathbb{R}^n$ hence (12*) is telling us geometrically that the length of a projection vector y is less than or equal to the length of x. $\square$

Proof. **3.** Let $x, y \in X$ where X is an inner product space. Let us take an orthonormal set $\{e_1, ..., e_n\}$ such that

$$e_1 = \frac{y}{\|y\|}$$

Then we see that

$$\begin{aligned}
\sum_{k=1}^{n} |\langle x, e_k \rangle|^2 &= |\langle x, e_1 \rangle|^2 + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2 \\
&= \left| \left\langle x, \frac{y}{\|y\|} \right\rangle \right|^2 + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2 \\
&= \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2
\end{aligned}$$

We note that

$$\frac{|\langle x, y \rangle|^2}{\|y\|^2} \leq \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2$$

Now, applying (12*) to x we get that

$$\frac{|\langle x, y \rangle|^2}{\|y\|^2} \leq \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Hence

$$|\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$$

And taking the square root we get Schwarz inequality

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

$\square$

Proof. **4.** Let us consider the orthonormal sequence

$$
\begin{aligned}
e_1 &= (0, 1, 0, 0, ...) \\
e_2 &= (0, 0, 1, 0, ...) \\
e_3 &= (0, 0, 0, 1, ...) \\
&\vdots
\end{aligned}
$$

And let us consider the sequence

$$x = (1, 0, 0, 0, ...) \in l^2$$

Then we see that

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2 = 0$$

But

$$\|x\|^2 = 1^2 + 0 + 0 + ... = 1$$

Therefore the strict inequality holds in this case

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2 = 0 < \|x\|^2 = 1$$

$\square$

Proof. **5.** Let (e_k) be an orthonormal sequence in an inner product space X, let also $x \in X$ and

$$y = \sum_{k=1}^{n} \alpha_k e_k \qquad \alpha_k = \langle x, e_k \rangle$$

We want to show that $x-y$ is orthogonal to the subspace $Y_n = \text{span}\{e_1, ..., e_n\}$. Let us consider some $y' \in Y_n$ defined as

$$y' = \sum_{k=1}^{n} \left\langle y', e_k \right\rangle e_k$$

Then

$$\begin{aligned}
\left\langle x - y, y' \right\rangle &= \left\langle x, y' \right\rangle - \left\langle y, y' \right\rangle \\
&= \left\langle x, \sum_{k=1}^{n} \left\langle y', e_k \right\rangle e_k \right\rangle - \left\langle \sum_{k=1}^{n} \langle x, e_k \rangle e_k, \sum_{j=1}^{n} \left\langle y', e_j \right\rangle e_j \right\rangle \\
&= \sum_{k=1}^{n} \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^{n} \langle x, e_k \rangle \left\langle e_k, \sum_{j=1}^{n} \left\langle y', e_j \right\rangle e_j \right\rangle \\
&= \sum_{k=1}^{n} \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^{n} \langle x, e_k \rangle \sum_{j=1}^{n} \overline{\langle y', e_j \rangle} \langle e_k, e_j \rangle
\end{aligned}$$

Since $\langle e_k, e_j \rangle = 1$ only if $k = j$ otherwise it's 0 then we can write that

$$\left\langle x - y, y' \right\rangle = \sum_{k=1}^{n} \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^{n} \langle x, e_k \rangle \overline{\langle y', e_k \rangle} = 0$$

Therefore since y' was arbitrary we see that $x - y$ is orthogonal to Y_n. $\square$

Proof. **6.** Let $\{e_1, ..., e_n\}$ be an orthonormal set in an inner product space X, where n is fixed. Let $x \in X$ and let $y = \beta_1 e_1 + ... + \beta_n e_n$. We want to show that $\|x - y\|$ is minimum if and only if $\beta_j = \langle x, e_j \rangle$ where $j = 1, ..., n$.

($\Rightarrow$) Suppose $\|x - y\|$ is minimum then we note that

$$x - y = x - \sum_{j=1}^{n} \beta_j e_j = \left(x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i \right) + \left(\sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{n} \beta_j e_j \right)$$

Then we see that

$$\begin{aligned}
&\left\langle x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i, \sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{n} \beta_j e_j \right\rangle \\
&= \left\langle x, \sum_{k=1}^{n} \langle x, e_k \rangle e_k \right\rangle - \left\langle x, \sum_{j=1}^{n} \beta_j e_j \right\rangle - \left\langle \sum_{i=1}^{n} \langle x, e_i \rangle e_i, \sum_{k=1}^{n} \langle x, e_k \rangle e_k \right\rangle \\
&\quad + \left\langle \sum_{i=1}^{n} \langle x, e_i \rangle e_i, \sum_{j=1}^{n} \beta_j e_j \right\rangle \\
&= \sum_{k=1}^{n} \overline{\langle x, e_k \rangle} \langle x, e_k \rangle - \sum_{j=1}^{n} \overline{\beta_j} \langle x, e_j \rangle - \sum_{i=1}^{n} \langle x, e_i \rangle \sum_{k=1}^{n} \overline{\langle x, e_k \rangle} \langle e_i, e_k \rangle \\
&\quad + \sum_{i=1}^{n} \langle x, e_i \rangle \sum_{j=1}^{n} \overline{\beta_j} \langle e_i, e_j \rangle \\
&= \sum_{k=1}^{n} \overline{\langle x, e_k \rangle} \langle x, e_k \rangle - \sum_{j=1}^{n} \overline{\beta_j} \langle x, e_j \rangle - \sum_{i=1}^{n} \langle x, e_i \rangle \overline{\langle x, e_i \rangle} + \sum_{i=1}^{n} \langle x, e_i \rangle \overline{\beta_i} \\
&= 0
\end{aligned}$$

So the vectors are orthogonal, then by the Pythagorean realation we have that

$$\|x - \sum_{j=1}^{n} \beta_j e_j\|^2 = \|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2 + \|\sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{n} \beta_j e_j\|^2$$

We see the last term is

$$\begin{aligned}
\|\sum_{k=1}^{n} (\langle x, e_k \rangle - \beta_k) e_k\|^2 &= \left\langle \sum_{k=1}^{n} (\langle x, e_k \rangle - \beta_k) e_k, \sum_{j=1}^{n} (\langle x, e_j \rangle - \beta_j) e_j \right\rangle \\
&= \sum_{k=1}^{n} (\langle x, e_k \rangle - \beta_k) \sum_{j=1}^{n} (\langle x, e_j \rangle - \beta_j) \langle e_k, e_j \rangle \\
&= \sum_{k=1}^{n} |\langle x, e_k \rangle - \beta_k|^2
\end{aligned}$$

Therefore since the first term, $\|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2$, is constant, and we said that $\|x - y\|^2$ is mininum then must be that

$$\sum_{k=1}^{n} |\langle x, e_k \rangle - \beta_k|^2 = 0$$

And that can only happen if $\beta_k = \langle x, e_k \rangle$ for all $k \in \{1, ..., n\}$.

($\Leftarrow$) Suppose now that $\beta_j = \langle x, e_j \rangle$ for all $j \in \{1, ..., n\}$ then

$$\|x - \sum_{j=1}^{n} \beta_j e_j\|^2 = \|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2 + \sum_{j=1}^{n} |\langle x, e_j \rangle - \beta_j|^2 = \|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2$$

Since $\|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2$ is a constant then $\|x - y\|$ cannot take a smaller value, i.e. it's the minimum when $\beta_j = \langle x, e_j \rangle$.

$\square$

Proof. **7.** Let (e_j) be any orthonormal sequence in an inner product space X and let $x, y \in X$.
Let us consider the Cauchy-Schwartz inequality over the sequences $(\langle x, e_k \rangle)$ and $(\langle y, e_k \rangle)$, then

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle \langle y, e_k \rangle | \leq \sqrt{\sum_{j=1}^{\infty} | \langle x, e_j \rangle |^2} \sqrt{\sum_{m=1}^{\infty} | \langle y, e_m \rangle |^2}$$

We know that $\sum_{j=1}^{\infty} | \langle x, e_j \rangle |^2 \leq \|x\|^2$ because of (12) hence

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle \langle y, e_k \rangle | \leq \sqrt{\|x\|^2} \sqrt{\|y\|^2} = \|x\| \|y\|$$

□

Proof. **8.** Let $x \in X$ where X is an inner product space and let n_m be the number of $\langle x, e_k \rangle$ such that $| \langle x, e_k \rangle | > 1/m$.
From equation (12) we know that

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2 \leq \|x\|^2$$

So we see that

$$\frac{n_m}{m^2} < \sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2 \leq \|x\|^2$$

Therefore, must be that

$$n_m < m^2 \|x\|^2$$

□

Proof. **9.** Let the sequence $(x_0, x_1, x_2, ...)$ where $x_j(t) = t^j$ on the interval $[-1, 1]$ where

$$\langle x, y \rangle = \int_{-1}^{1} x(t)y(t)\ dt$$

To orthonormalize the first three terms of the sequence we follow the Gram-Schmidt process. Let us compute first $\|x_0\|$ as follows

$$\|x_0\| = \sqrt{\langle x_0, x_0 \rangle} = \sqrt{\int_{-1}^{1} x_0(t)x_0(t)\ dt} = \sqrt{\int_{-1}^{1} t^0 t^0\ dt} = \sqrt{\int_{-1}^{1} dt} = \sqrt{2}$$

So the first element of our orthonormalized sequence is

$$e_0 = \frac{1}{\|x_0\|} x_0 = \frac{1}{\sqrt{2}} \cdot t^0 = \frac{1}{\sqrt{2}}$$

Now we compute v_1 as

$$\begin{aligned} v_1 &= x_1 - \langle x_1, e_0 \rangle e_0 \\ &= t - \frac{1}{\sqrt{2}} \cdot \int_{-1}^{1} t \cdot \frac{1}{\sqrt{2}}\ dt \\ &= t - \frac{1}{2} \int_{-1}^{1} t\ dt \\ &= t - \frac{1}{2} \left[\frac{t^2}{2} \right]_{-1}^{1} \\ &= t \end{aligned}$$

So the norm of v_1 is

$$\|v_1\| = \sqrt{\int_{-1}^{1} t^2\ dt} = \sqrt{\frac{2}{3}}$$

Then, the second element of the sequence is given by

$$e_1 = \frac{1}{\|v_1\|} v_1 = \sqrt{\frac{3}{2}} t$$

Finally, we compute v_2 as

$$\begin{aligned} v_2 &= x_2 - \langle x_2, e_1 \rangle e_1 - \langle x_2, e_0 \rangle e_0 \\ &= t^2 - \sqrt{\frac{3}{2}} t \cdot \int_{-1}^{1} t^2 \cdot \sqrt{\frac{3}{2}} t\ dt - \frac{1}{\sqrt{2}} \cdot \int_{-1}^{1} t^2 \cdot \frac{1}{\sqrt{2}}\ dt \\ &= t^2 - \frac{3}{2} t \cdot \int_{-1}^{1} t^3\ dt - \frac{1}{2} \int_{-1}^{1} t^2\ dt \\ &= t^2 - \frac{1}{3} \end{aligned}$$

Hence

$$
\begin{aligned}
\|v_2\| &= \sqrt{\int_{-1}^{1} \left(t^2 - \frac{1}{3}\right)^2 dt} \\
&= \sqrt{\int_{-1}^{1} \left(t^4 - \frac{2t^2}{3} + \frac{1}{9}\right) dt} \\
&= \sqrt{\frac{2}{5} - \frac{4}{9} + \frac{2}{9}} \\
&= \sqrt{\frac{8}{45}}
\end{aligned}
$$

so the third element of the sequence is

$$
e_2 = \frac{1}{\|v_2\|} v_2 = \sqrt{\frac{45}{8}} \left(t^2 - \frac{1}{3}\right)
$$

Just as a small sanity check we compute $\|e_2\|$ as follows

$$
\begin{aligned}
\|e_2\| &= \sqrt{\int_{-1}^{1} \frac{45}{8} \left(t^2 - \frac{1}{3}\right)^2 dt} \\
&= \sqrt{\frac{45}{8}} \sqrt{\int_{-1}^{1} \left(t^2 - \frac{1}{3}\right)^2 dt} \\
&= \sqrt{\frac{45}{8}} \|v_2\| \\
&= \sqrt{\frac{45}{8}} \sqrt{\frac{8}{45}} \\
&= 1
\end{aligned}
$$

$\square$

Proof. **1.** Suppose (6) converges to x i.e.

$$\sum_{k=1}^{\infty} \alpha_k e_k = x$$

Then by Theorem 3.5-2 we know that (7)

$$\sum_{k=1}^{\infty} |\alpha_k|^2$$

converges.
Let us take the squared norm of x then we have that

$$\|x\|^2 = \left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2$$

But since every $\alpha_k e_k$ is orthonormal then by the Pythagorean relation we have that

$$\left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2 = \sum_{k=1}^{\infty} \|\alpha_k e_k\|^2$$

Therefore

$$\|x\|^2 = \left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2 = \sum_{k=1}^{\infty} \|\alpha_k e_k\|^2 = \sum_{k=1}^{\infty} |\alpha_k|^2 \|e_k\|^2 = \sum_{k=1}^{\infty} |\alpha_k|^2$$

□

Proof. **3.** Let us consider the orthonormal sequence

$$\begin{aligned} e_1 &= (0, 1, 0, 0, ...) \\ e_2 &= (0, 0, 1, 0, ...) \\ e_3 &= (0, 0, 0, 1, ...) \\ &\vdots \end{aligned}$$

And let us consider the sequence

$$x = (1, 0, 0, 0, ...) \in l^2$$

Then we see that

$$\sum_{k=1}^{\infty} \langle x, e_k \rangle \, e_k = (0, 0, 0, 0, ...) \neq (1, 0, 0, 0, ...) = x$$

$\square$

Proof. **4.**
Let $(x_j) \subset X$ such that $\|x_1\| + \|x_2\| + ...$ converges.
We want to prove that (s_n) is Cauchy where $s_n = \sum_{i=1}^{n} x_i$. Taking $n', m' \in \mathbb{N}$ such that $n' < m'$, we see that

$$\left\| \sum_{i=1}^{n'} x_i - \sum_{j=1}^{m'} x_j \right\| = \left\| \sum_{i=n'+1}^{m'} x_i \right\| \leq \sum_{i=n'+1}^{m'} \|x_i\|$$

On the other hand, since $\sum_i \|x_i\|$ converges then, it's Cauchy, so, given $\epsilon > 0$ there is $N \in \mathbb{N}$ such that when $n, m \geq N$ we have that

$$\left| \sum_{i=1}^{n} \|x_i\| - \sum_{j=1}^{m} \|x_j\| \right| < \epsilon$$

We may assume that $n < m$, then

$$\left| \sum_{i=n+1}^{m} \|x_i\| \right| = \sum_{i=n+1}^{m} \|x_i\| < \epsilon$$

Hence, taking $n' = n$ and $m' = m$ we see that

$$\left\| \sum_{i=1}^{n} x_i - \sum_{j=1}^{m} x_j \right\| = \left\| \sum_{i=n+1}^{m} x_i \right\| \leq \sum_{i=n+1}^{m} \|x_i\| < \epsilon$$

Therefore (s_n) is Cauchy. $\square$

Proof. **5.**
From problem **4.** we know that if X is an inner product space and $\sum_i \|x_i\|$ converges then the sequence (s_n) with $s_n = \sum_{i=1}^{n} x_i$ is Cauchy.
If instead we take (s_n) in a Hilbert space H, the space is complete so all Cauchy sequences converge, hence the sequence of partial sums (s_n) converge to some limit s, so we have that

$$\sum_{i=1}^{\infty} x_i = \lim_{n\to\infty} \sum_{i=1}^{n} x_i = \lim_{n\to\infty} s_n = s$$

Which implies that $\sum_{i=1}^{\infty} x_i$ converges. $\square$

Proof. **6.**
Let (e_j) be an orthonormal sequence in Hilbert space H and let

$$x = \sum_{j=1}^{\infty} \alpha_j e_j \qquad y = \sum_{j=1}^{\infty} \beta_j e_j$$

Let us define the partial sums sequences

$$x_n = \sum_{j=1}^{n} \alpha_j e_j \qquad y_n = \sum_{j=1}^{n} \beta_j e_j$$

we see that $x_n \to x$ and $y_n \to y$ as $n \to \infty$.
Then

$$\langle x_n, y_n \rangle = \left\langle \sum_{j=1}^{n} \alpha_j e_j, \sum_{i=1}^{n} \beta_i e_i \right\rangle = \sum_{j=1}^{n} \alpha_j \sum_{i=1}^{n} \overline{\beta_i} \langle e_j, e_i \rangle = \sum_{j=1}^{n} \alpha_j \overline{\beta_j}$$

Because $x_n \to x$, $y_n \to y$ and the continuity of the inner product we have that $\langle x_n, y_n \rangle \to \langle x, y \rangle$. Therefore

$$\langle x, y \rangle = \lim_{n\to\infty} \langle x_n, y_n \rangle = \lim_{n\to\infty} \sum_{j=1}^{n} \alpha_j \overline{\beta_j} = \sum_{j=1}^{\infty} \alpha_j \overline{\beta_j}$$

$\square$

Proof. **7.**
Let (e_k) be an orthonormal sequence in a Hilbert space H and let $x \in H$. We want to show that

$$y = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

exists in H and that $x - y$ is orthogonal to every e_k.
Let us consider the partial sum sequence

$$y_n = \sum_{k=1}^{n} \langle x, e_k \rangle e_k$$

If we let $m < n$ then we see that

$$\begin{aligned}
\|y_n - y_m\|^2 &= \left\| \sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{m} \langle x, e_j \rangle e_j \right\|^2 \\
&= \left\| \sum_{k=m+1}^{n} \langle x, e_k \rangle e_k \right\|^2 \\
&= \sum_{k=m+1}^{n} \| \langle x, e_k \rangle e_k \|^2 \\
&= \sum_{k=m+1}^{n} | \langle x, e_k \rangle |^2
\end{aligned}$$

On the other hand, by Bessel inequality we know that $\sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2$ converges so the tail of the series $\sum_{k=m+1}^{n} | \langle x, e_k \rangle |^2$ must tend to 0 as $n, m \to \infty$.
Therefore $\|y_n - y_m\|^2 \to 0$ as $n, m \to \infty$ and hence (y_n) is Cauchy and since $(y_n) \in H$ then (y_n) converges. So

$$y = \lim_{n \to \infty} \sum_{k=1}^{n} \langle x, e_k \rangle e_k = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

exists in H.
Let us consider now the following

$$\begin{aligned}
\langle x - y_n, e_k \rangle &= \left\langle x - \sum_{j=1}^{n} \langle x, e_j \rangle e_j, e_k \right\rangle \\
&= \langle x, e_k \rangle - \left\langle \sum_{j=1}^{n} \langle x, e_j \rangle e_j, e_k \right\rangle \\
&= \langle x, e_k \rangle - \sum_{j=1}^{n} \langle x, e_j \rangle \langle e_j, e_k \rangle
\end{aligned}$$

Now taking the limit as $n \to \infty$ we get that

$$\begin{aligned}
\lim_{n\to\infty} \langle x - y_n, e_k \rangle &= \langle x, e_k \rangle - \lim_{n\to\infty} \sum_{j=1}^{n} \langle x, e_j \rangle \langle e_j, e_k \rangle \\
\langle x - y, e_k \rangle &= \langle x, e_k \rangle - \sum_{j=1}^{\infty} \langle x, e_j \rangle \langle e_j, e_k \rangle \\
\langle x - y, e_k \rangle &= \langle x, e_k \rangle - \langle x, e_k \rangle \\
\langle x - y, e_k \rangle &= 0
\end{aligned}$$

Where we used the continuity of the inner product, to show that as $n \to \infty$ we have that $\langle x - y_n, e_k \rangle \to \langle x - y, e_k \rangle$.
Therefore $x - y$ is orthogonal to every e_k.

$$\left\langle \sum_{j=1}^{\infty} \langle x, e_j \rangle e_j, e_k \right\rangle = \sum_{j=1}^{\infty} \langle x, e_j \rangle \langle e_j, e_k \rangle$$

$\square$

Proof. **8.**
Let (e_k) be an orthonormal sequence in a Hilbert space H and let M be $M = \operatorname{span}(e_k)$.

($\Rightarrow$) Let $x \in \bar{M}$ then x can be written as

$$x = \sum_{k=1}^{\infty} \alpha_k e_k$$

Where α_k are scalars. Since the summation converges to x then by Theorem 3.5-2 (b) we know the coefficients α_k are the Fourier coefficients $\langle x, e_k \rangle$ so x can be written as

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

($\Leftarrow$) Let $x \in H$ such that it can be written as

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k = \sum_{k=1}^{\infty} \alpha_k e_k$$

Then x is an infinite linear combination of the elements of the orthonormal sequence (e_k) so must be that $x \in \bar{M}$.

$\square$

Proof. **9.**
Let (e_n) and $(\tilde{e}_n)$ be orthonormal sequences in a Hilbert space H and let $M_1 = \text{span}(e_n)$ and $M_2 = \text{span}(\tilde{e}_n)$.

($\Rightarrow$) Let $\bar{M}_1 = \bar{M}_2$. By problem 8 we know that if $x \in \bar{M}_1$ and $x \in H$ then it can be written as

$$x = \sum_{m=1}^{\infty} \alpha_m e_m = \sum_{m=1}^{\infty} \langle x, e_m \rangle e_m$$

We note that $e_n \in \bar{M}_2$ then we can write it as

$$e_n = \sum_{m=1}^{\infty} \langle e_n, \tilde{e}_m \rangle \tilde{e}_m = \sum_{m=1}^{\infty} \alpha_{nm} \tilde{e}_m$$

Where we defined $\alpha_{nm} = \langle e_n, \tilde{e}_m \rangle$.
In the same way, since $\tilde{e}_n \in M_1$ then we can write it as

$$\tilde{e}_n = \sum_{m=1}^{\infty} \langle \tilde{e}_n, e_m \rangle e_m = \sum_{m=1}^{\infty} \overline{\langle e_m, \tilde{e}_n \rangle} e_m = \sum_{m=1}^{\infty} \bar{\alpha}_{mn} e_m$$

($\Leftarrow$) Let us suppose that

$$e_n = \sum_{m=1}^{\infty} \alpha_{nm} \tilde{e}_m \quad \text{and} \quad \tilde{e}_n = \sum_{m=1}^{\infty} \bar{\alpha}_{mn} e_m$$

Where $\alpha_{nm} = \langle e_n, \tilde{e}_m \rangle$.
We see that e_n and $\tilde{e}_n$ can be written in the form of equation (6) then the orhtonormal sequence (e_n) is both in $\bar{M}_1$ and $\bar{M}_2$ and the orhtonormal sequence $(\tilde{e}_n)$ is also in $\bar{M}_1$ and $\bar{M}_2$.
Let us take $x \in \bar{M}_1$ then by problem 8 we know that x can be written as

$$x = \sum_{m=1}^{\infty} \langle x, e_m \rangle e_m$$

But since the orthonormal sequence (e_n) is also in $\bar{M}_2$ then this implies that $x \in \bar{M}_2$ and since x is arbitrary then

$$\bar{M}_1 \subset \bar{M}_2$$

The opposite can be proven if $x \in \bar{M}_2$ so we get that

$$\bar{M}_1 = \bar{M}_2$$

□

3.6 - Total Orthonormal Sets and Sequences

Proof. **1.**
Let F be an orthonormal basis in an inner product space X. We know that

$$\overline{\text{span}(F)} = X$$

so if $x \in X$ can be represented as a linear combination of elements of F then $x \in \text{span}(F)$ but it could happen that $x \in X$ but it cannot be represented as a finite linear combination of elements in F.
For example, let us consider $X = l^2$ then if we take F as

$$F = \{(1, 0, 0, ...), (0, 1, 0, ...), (0, 0, 1, ...), ...\}$$

We see that $\overline{\text{span}(F)} = l^2$, also, the sequence $(1, 1/2, 1/3, ...) \in l^2$ but is not in $\text{span}(F)$ since it's infinite. $\square$

Proof. **2.**
Suppose the orthogonal dimension of a hilbert space H is finite, then there is a total orthonormal set T with finite number of elements in it, let us suppose this number is n, but by Lemma 3.4-2 we know that orthonormal sets are linealy independent so inside H we have a set of n linearly independent elements.
Let us suppose that there is, for example, one more element that is linearly independent with respect to the n previous elements, we want to arrive at a contradiction. So, in total we have $n + 1$ linearly independent elements, then, we can build an orthonormal set F of $n+1$ elements using the Gram-Schmidt process, since this new orthonormal set was built from the original total orthonormal set T then also, must be that $\text{span}(F)$ is dense in H, but for this to happen by Lemma 3.3-7 must be that $\text{span}(F)^{\perp} = \{0\}$ which implies that $\text{span}(T)^{\perp} \neq \{0\}$ in the first place.
This is a contradiction and must be that there cannot be more than n linearly independent sets if the orthogonal dimension of H is n.
Therefore, the vector space H is of dimension n.
Now, suppose H is considered as a vector space is of dimension n (finite), then there is a linearly independent set with n elements in it. We can obtain an orthonormal set of n elements by applying the Gram-Schmidt process and we observe that all elements of H are in the span of the orthonormal set, so the orthonormal set is total, hence, the orthogonal dimension of H is n. $\square$

Proof. **4.**
Let us consider first the case of a complex inner product space. From Parseval relation we note that

$$\|x-y\|^2 = \sum_k |\langle x-y, e_k\rangle|^2$$

Hence, from the LHS we see that

$$\|x-y\|^2 = \langle x-y, x-y\rangle = \langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle$$

And, from the RHS we have that

$$\begin{aligned}
\sum_k |\langle x-y, e_k\rangle|^2 &= \sum_k \langle x-y, e_k\rangle \overline{\langle x-y, e_k\rangle} \\
&= \sum_k (\langle x, e_k\rangle - \langle y, e_k\rangle)(\overline{\langle x, e_k\rangle} - \overline{\langle y, e_k\rangle}) \\
&= \sum_k \langle x, e_k\rangle \overline{\langle x, e_k\rangle} - \langle y, e_k\rangle \overline{\langle x, e_k\rangle} - \langle x, e_k\rangle \overline{\langle y, e_k\rangle} + \langle y, e_k\rangle \overline{\langle y, e_k\rangle} \\
&= \sum_k |\langle x, e_k\rangle|^2 - \langle y, e_k\rangle \overline{\langle x, e_k\rangle} - \langle x, e_k\rangle \overline{\langle y, e_k\rangle} + |\langle y, e_k\rangle|^2 \\
&= \langle x,x\rangle + \langle y,y\rangle - \sum_k \langle y, e_k\rangle \overline{\langle x, e_k\rangle} + \langle x, e_k\rangle \overline{\langle y, e_k\rangle}
\end{aligned}$$

Joining both results we see that

$$\langle x,y\rangle + \langle y,x\rangle = \sum_k \langle y, e_k\rangle \overline{\langle x, e_k\rangle} + \langle x, e_k\rangle \overline{\langle y, e_k\rangle}$$

Now, let us consider

$$\|x-iy\|^2 = \sum_k |\langle x-iy, e_k\rangle|^2$$

From the LHS we see that

$$\begin{aligned}
\|x-iy\|^2 &= \langle x-iy, x-iy\rangle \\
&= \langle x,x\rangle - \langle x,iy\rangle - \langle iy,x\rangle + \langle iy,iy\rangle \\
&= \langle x,x\rangle + i\langle x,y\rangle - i\langle y,x\rangle + \langle y,y\rangle
\end{aligned}$$

And, from the RHS we have that

$$\begin{aligned}
\sum_k |\langle x-iy, e_k\rangle|^2 &= \sum_k \langle x-iy, e_k\rangle \overline{\langle x-iy, e_k\rangle} \\
&= \sum_k (\langle x, e_k\rangle - i\langle y, e_k\rangle)(\overline{\langle x, e_k\rangle} + i\overline{\langle y, e_k\rangle}) \\
&= \sum_k \langle x, e_k\rangle \overline{\langle x, e_k\rangle} - i\langle y, e_k\rangle \overline{\langle x, e_k\rangle} + i\langle x, e_k\rangle \overline{\langle y, e_k\rangle} + \langle y, e_k\rangle \overline{\langle y, e_k\rangle} \\
&= \sum_k |\langle x, e_k\rangle|^2 - i\langle y, e_k\rangle \overline{\langle x, e_k\rangle} + i\langle x, e_k\rangle \overline{\langle y, e_k\rangle} + |\langle y, e_k\rangle|^2 \\
&= \langle x,x\rangle + \langle y,y\rangle + i\sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle} - \langle y, e_k\rangle \overline{\langle x, e_k\rangle}
\end{aligned}$$

Again, joining both results we see that

$$\langle x, y \rangle - \langle y, x \rangle = \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle} - \langle y, e_k \rangle \overline{\langle x, e_k \rangle}$$

Adding both result equations we get that

$$\begin{aligned}&\langle x, y \rangle + \langle y, x \rangle + \langle x, y \rangle - \langle y, x \rangle \\ &= \sum_k \langle y, e_k \rangle \overline{\langle x, e_k \rangle} + \langle x, e_k \rangle \overline{\langle y, e_k \rangle} + \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle} - \langle y, e_k \rangle \overline{\langle x, e_k \rangle}\end{aligned}$$

Hence

$$\langle x, y \rangle = \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle}$$

Let us consider now the case of a real inner product space. From the polarization identity we know that

$$\langle x, y \rangle = \frac{1}{2}(\|x\|^2 + \|y\|^2 - \|x - y\|^2)$$

Hence

$$\begin{aligned}\langle x, y \rangle &= \frac{1}{2}(\langle x, x \rangle + \langle y, y \rangle - \langle x, x \rangle - \langle y, y \rangle + \sum_k \langle y, e_k \rangle \langle x, e_k \rangle + \langle x, e_k \rangle \langle y, e_k \rangle) \\ &= \frac{1}{2}(2 \sum_k \langle x, e_k \rangle \langle y, e_k \rangle) \\ &= \sum_k \langle x, e_k \rangle \langle y, e_k \rangle\end{aligned}$$

Where we used the previous result we got for $\|x - y\|^2$ and that $\langle x, e_k \rangle = \langle e_k, x \rangle$ in a real inner product space. $\square$

Proof. **5.**
Let (e_κ), with $\kappa \in I$ be an orthonormal family in a Hilbert space H.

($\Rightarrow$) Let us suppose (e_κ) is total. Since Parseval relation

$$\sum_k |\langle x, e_k\rangle|^2 = \|x\|^2$$

Holds for any total orthonormal set M in a Hilbert space H and we derived the equation

$$\langle x, y\rangle = \sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle}$$

Using the Parseval relation then it must also hold for the orthonormal family (e_κ).

($\Leftarrow$) Let us suppose that for every $x, y \in H$ the following equation holds

$$\langle x, y\rangle = \sum_\kappa \langle x, e_\kappa\rangle \overline{\langle y, e_\kappa\rangle}$$

For the orthonormal family (e_κ). Taking $y = x$ we get that

$$\|x\|^2 = \langle x, x\rangle = \sum_\kappa \langle x, e_\kappa\rangle \overline{\langle x, e_\kappa\rangle} = \sum_\kappa |\langle x, e_\kappa\rangle|^2$$

Then Parseval relation also holds and by Theorem 3.6-3 then must be that the orthonormal family (e_κ) is total.

$\square$

Proof. **6.**
Let H be a separable Hilbert space and $M = \{x_1, x_2, x_3, ...\}$ a countable dense subset of H. Then, let us select

$$e_1 = \frac{x_1}{\|x_1\|}$$

To select e_2 we check if x_2 is in $\mathrm{span}(e_1)$, if it is there then we drop it, if it's not then we define $v_2 = x_2 - \langle x_2, e_1 \rangle e_1$ and we select

$$e_2 = \frac{v_2}{\|v_2\|}$$

To select e_3 we check if x_3 is in $\mathrm{span}(\{e_1, e_2\})$, if it is there then we drop it, if it's not then we define $v_3 = x_3 - \langle x_3, e_1 \rangle e_1 - \langle x_3, e_2 \rangle e_2$ and we select

$$e_3 = \frac{v_3}{\|v_3\|}$$

i.e. we follow the Gram-Schmidt process dropping some elements if necessary, to get an orthonormal sequence $(e_j) \subseteq M$.
Since the elements we dropped are in some span then they can be written as a linear combination of elements of (e_j), so $M \subseteq \mathrm{span}(e_j)$ but also we know that M is a dense subset of H so must be that $\mathrm{span}(e_j)$ is also dense in H. Therefore (e_j) is a total orthonormal sequence obtained from M. $\square$

Proof. **7.**
Let H be a separable Hilbert space, then there is M a countable dense subset in H. As we showed before, using the Gram-Schmidt process we can find a total orthonormal sequence in H. $\square$

Proof. **8.**
Let $F = (e_j)$ be an orthonormal sequence in a separable Hilbert space H.
We know there exists $D = \{d_1, d_2, d_3, ...\}$, a countable dense subset of H, because H is separable.
Let us take d_1, if d_1 is linearly independent from the rest of the elements in D we keep it.
Next, we compute

$$v_1 = d_1 - \sum_{k=1}^{\infty} \langle d_1, e_k \rangle e_k$$

and

$$x_1 = \frac{v_1}{\|v_1\|}$$

i.e. we follow the Gram-Schmidt process to orthonormalize d_1.
Finally, we add x_1 to a new orthogonal sequence $\tilde{F}$ defined as

$$\tilde{F} = \{x_1, x_2, ..., e_1, e_2, ...\}$$

To get each x_i we follow the same process for each d_i. If d_i is not linearly independent we drop it and continue.
Following this process we see that $\tilde{F}$ is an orthonormal sequence, and since we started from a countable dense subset D, then $\operatorname{span}(\tilde{F})$ must also be dense in H, since we only dropped the elements that were linearly dependent to the other elements.
Therefore $\tilde{F}$ is a total orthonormal sequence that contains F. $\square$

Proof. **9.**
Let M be a total set in an inner product space X and let $\langle v, x \rangle = \langle w, x \rangle$ for all $x \in M$. We note that

$$\begin{aligned} \langle v, x \rangle &= \langle w, x \rangle \\ \langle v, x \rangle - \langle w, x \rangle &= 0 \\ \langle v - w, x \rangle &= 0 \end{aligned}$$

This implies that $v - w \perp x$ but because Theorem 3.6-2, there is no element in X which is orthogonal to every element in M so must be that $v - w = 0$ and therefore $v = w$. $\square$

Proof. **10.**
Let M be a subset of a Hilbert space H and let $v, w \in H$. Suppose that $\langle v, x\rangle = \langle w, x\rangle$ for all $x \in M$ implies that $v = w$.
If M is not total then there is some $y \in H$, $y \neq 0$ such that $y \perp M$, we want to arrive at a contradiction.
Then $\langle y, x\rangle = 0$, but also we know that $\langle 0, x\rangle = 0$ so $\langle y, x\rangle = \langle 0, x\rangle$ hence $y = 0$ but y was different from 0, a contradiction.
Therefore M is total in H. $\square$

3.7 - Legendre, Hermite and Laguerre Polynomials

Proof. **1.**
The Legendre differential equation is

$$(1-t^2)P_n'' - 2tP_n' + n(n+1)P_n = 0$$

We note that applying the product rule to $[(1-t^2)P_n']'$ gives us

$$[(1-t^2)P_n']' = (1-t^2)'P_n' + (1-t^2)P_n'' = (1-t^2)P_n'' - 2tP_n'$$

Then we can write the Legendre differential equation as

$$[(1-t^2)P_n']' = -n(n+1)P_n$$

Multiplying this equation by P_m, where $m \neq n$, gives us

$$P_m[(1-t^2)P_n']' = -n(n+1)P_nP_m$$

The corresponding equation for P_m multiplied by $-P_n$ is

$$-P_n[(1-t^2)P_m']' = m(m+1)P_mP_n$$

Then, adding the equation and integrating from -1 to 1 we see that

$$\begin{aligned}\int_{-1}^{1} P_m[(1-t^2)P_n']' - P_n[(1-t^2)P_m']'\ dt &= \int_{-1}^{1} m(m+1)P_mP_n - n(n+1)P_nP_m\ dt \\ &= [m(m+1) - n(n+1)]\int_{-1}^{1} P_nP_m\ dt\end{aligned}$$

Let us integrate $\int_{-1}^{1} P_m[(1-t^2)P_n']'\ dt$ by parts

$$\begin{aligned}\int_{-1}^{1} P_m[(1-t^2)P_n']'\ dt &= P_m[(1-t^2)P_n']\Big|_{-1}^{1} - \int_{-1}^{1} P_m'[(1-t^2)P_n']\ dt \\ &= 0 - \int_{-1}^{1}(1-t^2)P_n'P_m'\ dt\end{aligned}$$

In the same way for $\int_{-1}^{1} P_n[(1-t^2)P_m']'\ dt$ we get that

$$\begin{aligned}\int_{-1}^{1} P_n[(1-t^2)P_m']'\ dt &= P_n[(1-t^2)P_m']\Big|_{-1}^{1} - \int_{-1}^{1} P_n'[(1-t^2)P_m']\ dt \\ &= 0 - \int_{-1}^{1}(1-t^2)P_n'P_m'\ dt\end{aligned}$$

Therefore

$$\begin{aligned}&\int_{-1}^{1} P_m[(1-t^2)P_n']' - P_n[(1-t^2)P_m']'\ dt \\ &= -\int_{-1}^{1}(1-t^2)P_n'P_m'\ dt + \int_{-1}^{1}(1-t^2)P_n'P_m'\ dt \\ &= 0\end{aligned}$$

So

$$\langle P_n, P_m \rangle = \int_{-1}^{1} P_n P_m \; dt = 0$$

Hence (P_n) is an orthogonal sequence in the space $L^2[-1, 1]$.

$\square$

Proof. **2.**
The binomial theorem states that

$$(a+b)^n = \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} a^{n-j} b^j$$

So in this case we can apply it to $(t^2-1)^n$ as follows

$$(t^2-1)^n = \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} t^{2n-2j} (-1)^j$$

Let us take the nth derivative of t^{2n-2j}, we get that

$$\begin{aligned}
\frac{\mathrm{d}^n}{\mathrm{d}t^n} t^{2n-2j} &= (2n-2j) \frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}} t^{2n-2j-1} \\
&= (2n-2j) \cdot (2n-2j-1) \frac{\mathrm{d}^{n-2}}{\mathrm{d}t^{n-2}} t^{2n-2j-2} \\
&\ \vdots \\
&= [(2n-2j) \cdot (2n-2j-1) \cdot \ldots \cdot (2n-2j-n)] t^{2n-2j-n} \\
&= \frac{(2n-2j)!}{(n-2j)!} t^{n-2j}
\end{aligned}$$

Hence

$$\frac{\mathrm{d}^n}{\mathrm{d}t^n} \left(t^2-1\right)^n = \sum_{j=0}^{n} (-1)^j \frac{n!}{j!(n-j)!} \frac{(2n-2j)!}{(n-2j)!} t^{n-2j}$$

Therefore

$$\begin{aligned}
P_n(t) &= \frac{1}{2^n n!} \frac{\mathrm{d}^n}{\mathrm{d}t^n} [(t^2-1)^n] \\
&= \frac{1}{2^n n!} \sum_{j=0}^{n} (-1)^j \frac{n!}{j!(n-j)!} \frac{(2n-2j)!}{(n-2j)!} t^{n-2j} \\
&= \sum_{j=0}^{n} (-1)^j \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j)!} t^{n-2j}
\end{aligned}$$

□

Proof. **3.**
Let us consider the LHS as a binomial series

$$\frac{1}{\sqrt{1-2tw+w^2}} = (1-2tw+w^2)^{-1/2} = \sum_{k=0}^{\infty} \binom{-1/2}{k} (-2tw+w^2)^k$$

Now applying the binomial theorem we get that

$$\begin{aligned}
\frac{1}{\sqrt{1-2tw+w^2}} &= \sum_{k=0}^{\infty} \binom{-1/2}{k} \sum_{j=0}^{k} \binom{k}{j} (-2tw)^{k-j} w^{2j} \\
&= \sum_{k=0}^{\infty} \binom{-1/2}{k} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} 2^{k-j} t^{k-j} w^{2j+k-j} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k} (-1)^{k-j} \frac{(-1/2)(-1/2-1)...(-1/2-k+1)}{k!} \frac{k!}{j!(k-j)!} 2^{k-j} t^{k-j} w^{j+k} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k} (-1)^{k-j} \frac{2^k(-1/2)(-1/2-1)...(-1/2-k+1)}{2^k j!(k-j)!} 2^{k-j} t^{k-j} w^{j+k} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k} (-1)^{k-j} \frac{2^k \prod_{i=0}^{k-1} -(2i+1)/2}{2^j j!(k-j)!} t^{k-j} w^{j+k} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k} (-1)^{k-j} \frac{\prod_{i=0}^{k-1} -(2i+1)}{2^j j!(k-j)!} t^{k-j} w^{j+k} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k} (-1)^{k-j} \frac{(-1)^k \prod_{i=0}^{k-1} (2i+1)}{2^j j!(k-j)!} t^{k-j} w^{j+k} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k} (-1)^{2k-j} \frac{2k!}{2^{k+j} k! j!(k-j)!} t^{k-j} w^{j+k} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k} (-1)^{j} \frac{2k!}{2^{k+j} k! j!(k-j)!} t^{k-j} w^{j+k}
\end{aligned}$$

Where we used in the last step that $(-1)^{2k-j} = (-1)^{2k}(-1)^{-j} = 1 \cdot (-1)^j$. Applying the formula $\sum_{k=0}^{\infty} \sum_{j=0}^{k} C_{j,k} t^{k+j} = \sum_{k=0}^{\infty} \sum_{j=0}^{k/2} C_{j,k-j} t^k$ we get that

$$\begin{aligned}
\frac{1}{\sqrt{1-2tw+w^2}} &= \sum_{k=0}^{\infty} \sum_{j=0}^{k/2} (-1)^j \frac{(2(k-j))!}{2^{k-j+j}(k-j)!j!(k-j-j)!} t^{(k-j)-j} w^{j+k-j} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{k/2} (-1)^j \frac{(2k-2j)!}{2^k(k-j)!j!(k-2j)!} t^{k-2j} w^k \\
&= \sum_{k=0}^{\infty} P_k(t) w^k
\end{aligned}$$

$\square$

Proof. **4.**
Let us consider

$$\frac{1}{r} = \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\theta}}$$

but let us write it as

$$\frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\theta}} = \frac{1}{r_2} \frac{1}{\sqrt{1 + (r_1/r_2)^2 - 2(r_1/r_2)\cos\theta}}$$

Then taking $w = r_1/r_2$ and $t = \cos\theta$ the RHS has the form $1/r_2\sqrt{1 - 2tw + w^2}$ and hence using the result of the previous problem we get that

$$\begin{aligned}
&\frac{1}{r_2} \frac{1}{\sqrt{1 - 2(r_1/r_2)\cos\theta + (r_1/r_2)^2}} = \\
&\quad = \frac{1}{r_2} \frac{1}{\sqrt{1 - 2wt + w^2}} \\
&\quad = \frac{1}{r_2} \sum_{n=0}^{\infty} P_n(t) w^n \\
&\quad = \frac{1}{r_2} \sum_{n=0}^{\infty} P_n(\cos\theta) \left(\frac{r_1}{r_2}\right)^n
\end{aligned}$$

□

Proof. **5.**
Let us consider $x(t) = \sum_{k=0}^{\infty} c_k t^k$, we note that

$$x(t)' = \sum_{k=1}^{\infty} c_k k t^{k-1}$$
$$x(t)'' = \sum_{k=2}^{\infty} c_k k(k-1) t^{k-2}$$

Replacing this into Legendre's differential equation gives us

$$\begin{aligned}(1-t^2)x(t)'' - 2tx(t)' + n(n+1)x(t) = 0\\
(1-t^2)\sum_{k=2}^{\infty} c_k k(k-1)t^{k-2} - 2t\sum_{k=1}^{\infty} c_k k t^{k-1} + n(n+1)\sum_{k=0}^{\infty} c_k t^k = 0\\
\sum_{k=2}^{\infty} c_k k(k-1)t^{k-2} - \sum_{k=2}^{\infty} c_k k(k-1)t^k - \sum_{k=1}^{\infty} 2c_k k t^k + \sum_{k=0}^{\infty} n(n+1)c_k t^k = 0\\
\sum_{k=0}^{\infty} c_{k+2}(k+1)(k+2)t^k - \sum_{k=2}^{\infty} c_k k(k-1)t^k - \sum_{k=1}^{\infty} 2c_k k t^k + \sum_{k=0}^{\infty} n(n+1)c_k t^k = 0\end{aligned}$$

In the second term for $k = 0$ and 1 the terms are 0, so we can set the sum to start at $k = 0$, the same applies for the third term, hence

$$\sum_{k=0}^{\infty} [c_{k+2}(k+1)(k+2) - c_k k(k-1) - 2c_k k + n(n+1)c_k]t^k = 0$$

Since the sum is equal to 0 then must be that each term is 0, so we get that

$$\begin{aligned}c_{k+2}(k+1)(k+2) = [k(k-1) + 2k - n(n+1)]c_k\\
c_{k+2} = \frac{k(k-1) + 2k - n(n+1)}{(k+1)(k+2)} c_k\end{aligned}$$

We see that

$$\begin{aligned}
c_2 &= -\frac{n(n+1)}{2} c_0 && k = 0\\
c_3 &= \frac{2 - n(n+1)}{6} c_1 = -\frac{(n-1)(n+2)}{6} c_1 && k = 1\\
c_4 &= \frac{6 - n(n+1)}{12} c_2 = -\frac{(n-2)(n+3)}{12} c_2 = && k = 2\\
&= \frac{(n-2)(n+3)}{12} \frac{n(n+1)}{2} c_0 = \frac{(n-2)(n+3)n(n+1)}{24} c_0\\
&...
\end{aligned}$$

Then we can write $x(t)$ as

$$\begin{aligned}x(t) = c_0\left[1 - \frac{n(n+1)}{2!}t^2 + \frac{(n-2)(n+3)n(n+1)}{4!}t^4 + ...\right] +\\
+ c_1\left[t - \frac{(n-1)(n+2)}{3!}t^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}t^5 + ...\right]\end{aligned}$$

or defining $x_1(t)$ an $x_2(t)$ as

$$x_1(t) = 1 - \frac{n(n+1)}{2!}t^2 + \frac{(n-2)(n+3)n(n+1)}{4!}t^4 + ...$$
$$x_2(t) = t - \frac{(n-1)(n+2)}{3!}t^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}t^5 + ...$$

We can write it as $x(t) = c_0 x_1(t) + c_1 x_2(t)$.
Finally, we note that when n is even, then the term containing t^{n+2} a $x_1(t)$ becomes 0, and since the coefficients are recursive, all subsequent coefficients become 0. The same thing happens when n is odd, so $x_2(t)$ becomes also a polynomial.
On the other hand, if we assume that

$$c_{n-2j} = (-1)^j \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j)!}$$

Then $c_{n-2(j-1)} = c_{n-2j+2}$ must be

$$c_{n-2j+2} = (-1)^{j-1} \frac{(2n-2j+2)!}{2^n (j-1)!(n-j+1)!(n-2j+2)!}$$

But also by the recurrence relation, setting $k = n-2j$ we have that

$$\begin{aligned}
c_{n-2j+2} &= \frac{(n-2j)(n-2j-1)+2(n-2j)-n(n+1)}{(n-2j+1)(n-2j+2)} c_{n-2j} \\
&= (-1)^j \frac{(n-2j)(n-2j-1)+2(n-2j)-n(n+1)}{(n-2j+1)(n-2j+2)} \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j)!} \\
&= (-1)^j [(n-2j)(n-2j+1)-n(n+1)] \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j+2)!} \\
&= (-1)^j [n^2-2jn+n-2jn+4j^2-2j-n(n+1)] \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j+2)!} \\
&= (-1)^j [-4jn+4j^2-2j] \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j+2)!} \\
&= (-1)^j [-2j(2n-2j+1)] \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j+2)!} \\
&= (-1)^{j-1} \frac{2j!}{(j-1)!} \frac{(2n-2j+1)!}{2^n j!(n-j)!(n-2j+2)!} \\
&= (-1)^{j-1} \frac{2(n-j+1)}{(n-j+1)} \frac{(2n-2j+1)!}{2^n (j-1)!(n-j)!(n-2j+2)!} \\
&= (-1)^{j-1} \frac{(2n-2j+2)!}{2^n (j-1)!(n-j+1)!(n-2j+2)!}
\end{aligned}$$

Therefore the coefficients of x_1 and x_2 match the coefficients of the Legendre polynomials, if we set correctly the c_0 and c_1. $\square$

Proof. **6.**
Let us consider the LHS as

$$e^{2wt-w^2} = \sum_{n=0}^{\infty} \frac{1}{n!}(2wt - w^2)^n$$

Applying the binomial theorem we get that

$$\begin{aligned}
e^{2wt-w^2} &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} (2wt)^{n-j}(-w^2)^j \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} 2^{n-j} w^{n-j} t^{n-j} (-1)^j w^{2j} \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \left[n! \sum_{j=0}^{n} (-1)^j \frac{2^{n-j}}{j!(n-j)!} t^{n-j} w^{n+j} \right]
\end{aligned}$$

Also, we know that $\sum_{n=0}^{\infty} \sum_{j=0}^{n} C_{j,n} t^{n+j} = \sum_{n=0}^{\infty} \sum_{j=0}^{[n/2]} C_{j,n-j} t^n$.
Therefore

$$e^{2wt-w^2} = \sum_{n=0}^{\infty} \frac{1}{n!} \left[n! \sum_{j=0}^{[n/2]} (-1)^j \frac{2^{n-2j}}{j!(n-2j)!} t^{n-2j} w^n \right] = \sum_{n=0}^{\infty} \frac{1}{n!} H_n(t) w^n$$

□

Proof. **7.**
We know that

$$H_n(t) = (-1)^n e^{t^2} \frac{\mathrm{d}^n}{\mathrm{d}t^n}\left(e^{-t^2}\right)$$

So we see that

$$\begin{aligned}
H_n'(t) &= (-1)^n \left[2te^{t^2} \frac{\mathrm{d}^n}{\mathrm{d}t^n}\left(e^{-t^2}\right) + e^{t^2} \frac{\mathrm{d}^{n+1}}{\mathrm{d}t^{n+1}}\left(e^{-t^2}\right)\right] \\
-H_n'(t) &= -2t(-1)^n e^{t^2} \frac{\mathrm{d}^n}{\mathrm{d}t^n}\left(e^{-t^2}\right) - (-1)^n e^{t^2} \frac{\mathrm{d}^{n+1}}{\mathrm{d}t^{n+1}}\left(e^{-t^2}\right) \\
-H_n'(t) &= -2t(-1)^n e^{t^2} \frac{\mathrm{d}^n}{\mathrm{d}t^n}\left(e^{-t^2}\right) + (-1)^{n+1} e^{t^2} \frac{\mathrm{d}^{n+1}}{\mathrm{d}t^{n+1}}\left(e^{-t^2}\right)
\end{aligned}$$

Now, replacing the definition of $H_n(t)$ and that

$$H_{n+1}(t) = (-1)^{n+1} e^{t^2} \frac{\mathrm{d}^{n+1}}{\mathrm{d}t^{n+1}}\left(e^{-t^2}\right)$$

We get that

$$-H_n'(t) = -2tH_n(t) + H_{n+1}(t)$$

Therefore

$$H_{n+1}(t) = 2tH_n(t) - H_n'(t)$$

□

Proof. **8.**
Let us differentiate the generating function as follows

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{2wt-w^2}\right) = 2we^{2wt-w^2} = \sum_{n=1}^{\infty} \frac{1}{n!} H_n'(t) w^n$$

The first term of the sum becomes 0 since $H_0(t) = 1$, so the sum starts at 1. Then we have that

$$\begin{aligned}
2w \sum_{n=0}^{\infty} \frac{1}{n!} H_n(t) w^n &= \sum_{n=1}^{\infty} \frac{1}{n!} H_n'(t) w^n \\
2w \sum_{n=1}^{\infty} \frac{1}{(n-1)!} H_{n-1}(t) w^{n-1} &= \sum_{n=1}^{\infty} \frac{1}{n!} H_n'(t) w^n \\
2 \sum_{n=1}^{\infty} \frac{1}{n!} n H_{n-1}(t) w^n &= \sum_{n=1}^{\infty} \frac{1}{n!} H_n'(t) w^n \\
\sum_{n=1}^{\infty} \frac{1}{n!} [2nH_{n-1}(t) - H_n'(t)] w^n = 0 &
\end{aligned}$$

Where in the second step we changed the LHS sum to start from 1. Since the sum is equal to 0 then must be that each term is 0, hence

$$H_n'(t) = 2nH_{n-1}(t)$$

The Hermite differential equation states that

$$H_n'' - 2tH_n' + 2nH_n = 0$$

From the equation we derived and the result of Problem 7 with $n-1$ we see that

$$H_n'' = 2nH_{n-1}' = 2n(2tH_{n-1} - H_n)$$

Now, replacing in the Hermite differential equation we get that

$$\begin{aligned}
H_n'' - 2tH_n' + 2nH_n &= 2n(2tH_{n-1} - H_n) - 2t(2nH_{n-1}) + 2nH_n \\
&= 4ntH_{n-1} - 2nH_n - 4tnH_{n-1} + 2nH_n \\
&= 0
\end{aligned}$$

Therefore H_n satisfies the Hermite differential equation. $\square$

Proof. **9.**
In the asymptotic case the differential equation $y'' + (2n + 1 - t^2)y = 0$ becomes $y'' = t^2 y$ for which we know that the solution is $y(t) = e^{-t^2/2}$ so we may assume that the solution to the original differential equation is of the form $y(t) = h(t)e^{-t^2/2}$, replacing this solution into the differential equation we have that

$$\begin{aligned}
\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + (2n+1-t^2)y &= 0 \\
\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(h(t)e^{-t^2/2}\right) + (2n+1-t^2)h(t)e^{-t^2/2} &= 0 \\
e^{-t^2/2}(h(t)'' - 2th(t)' + (t^2-1)h(t)) + (2n+1-t^2)h(t)e^{-t^2/2} &= 0 \\
h(t)'' - 2th(t)' + (t^2-1)h(t) + 2nh(t) - (t^2-1)h(t) &= 0 \\
h(t)'' - 2th(t)' + 2nh(t) &= 0
\end{aligned}$$

We see that the differential equation is of the form of the Hermite differential equation, so H_n is a solution to the equation.
Therefore $H_n(t)e^{-t^2/2}$ is a solution to the original differential equation. □

Proof. **10.**
Let us take the derivative of $e^{-t^2}H_n'$ as follows

$$(e^{-t^2}H_n')' = -2te^{-t^2}H_n' + e^{-t^2}H_n''$$

Also, from Problem 8 we know that $H_n' = 2nH_{n-1}$ so

$$H_n'' = 2nH_{n-1}' = 2n(2tH_{n-1} - H_n)$$

Where we used that $H_{n-1}' = 2tH_{n-1} - H_n$ from Problem 7.
Hence replacing

$$\begin{aligned}(e^{-t^2}H_n')' &= -4nte^{-t^2}H_{n-1} + 4nte^{-t^2}H_{n-1} - 2ne^{-t^2}H_n \\ &= -2ne^{-t^2}H_n\end{aligned}$$

From the equation we just derived if we let $n \neq m$ and following the same method as in Problem 1, we can write that

$$\begin{aligned}H_m(e^{-t^2}H_n')' - H_n(e^{-t^2}H_m')' &= -2ne^{-t^2}H_nH_m + 2me^{-t^2}H_nH_m \\ &= 2e^{-t^2}(m-n)H_nH_m\end{aligned}$$

The (7a) functions state that

$$e_n = A_ne^{-t^2/2}H_n$$

where $A_n = 1/(2^n n!\sqrt{\pi})^{1/2}$.
We want to check that the functions are orthogonal on $\mathbb{R}$ so we need to integrate e_ne_m from $-\infty$ to ∞

$$A_nA_m\int_{-\infty}^{\infty} e^{-t^2/2}H_ne^{-t^2/2}H_m\ dt = A_nA_m\int_{-\infty}^{\infty} e^{-t^2}H_nH_m\ dt$$

We note that this is the same as integrating $H_m(e^{-t^2}H_n')' - H_n(e^{-t^2}H_m')'$ times a multiplicative constant, so integrating the first term by parts we see that

$$\begin{aligned}\int_{-\infty}^{\infty}(e^{-t^2}H_n')'H_m\ dt &= e^{-t^2}H_n'H_m\Big|_{t=-\infty}^{t=\infty} - \int_{-\infty}^{\infty} e^{-t^2}H_n'H_m'\ dt \\ &= 0 - \int_{-\infty}^{\infty} e^{-t^2}H_n'H_m'\ dt\end{aligned}$$

In hte same way, integrating by parts the second term we get that

$$\begin{aligned}-\int_{-\infty}^{\infty}(e^{-t^2}H_m')'H_n\ dt &= -e^{-t^2}H_m'H_n\Big|_{t=-\infty}^{t=\infty} + \int_{-\infty}^{\infty} e^{-t^2}H_m'H_n'\ dt \\ &= 0 + \int_{-\infty}^{\infty} e^{-t^2}H_n'H_m'\ dt\end{aligned}$$

Therefore

$$\int_{-\infty}^{\infty} H_m(e^{-t^2}H_n')' - H_n(e^{-t^2}H_m')' \; dt = 2(m-n)\int_{-\infty}^{\infty} e^{-t^2}H_nH_m \; dt = 0$$

Hence

$$\langle e_n, e_m \rangle = A_nA_m \int_{-\infty}^{\infty} e^{-t^2}H_nH_m \; dt = 0$$

Which impliest that the functions $e_n(t)$ are orthogonal. $\square$

Proof. **11.**
Let us consider the LHS of the equation as follows

$$\begin{aligned}
\frac{1}{1-w}e^{-tw/(1-w)} &= \frac{1}{1-w}\sum_{n=0}^{\infty}\frac{1}{n!}\left(-\frac{tw}{1-w}\right)^n \\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\frac{(-1)^n w^n}{(1-w)^{n+1}}t^n
\end{aligned}$$

Using the series expansion for $1/(1-w)^{n+1}$ we get that

$$\begin{aligned}
\frac{1}{1-w}e^{-tw/(1-w)} &= \sum_{n=0}^{\infty}\frac{1}{n!}(-1)^n w^n t^n \sum_{k=0}^{\infty}\binom{k+n}{k}w^k \\
&= \sum_{n=0}^{\infty}\sum_{k=0}^{\infty}\frac{(-1)^n}{n!}\binom{k+n}{k}w^{n+k}t^n \\
&= \sum_{n=0}^{\infty}\sum_{j=n}^{\infty}\frac{(-1)^n}{n!}\binom{j}{n}w^j t^n
\end{aligned}$$

Where in the last step we changed variables so $j = k+n$ and hence the sum starts at $j = n$.
If $n = 0$ then j can go from 0 to ∞, if $n = 1$ then j can go from 1 to ∞ and so on. If we swap the summation signs for $j = 0$ then n can only be 0, for $j = 1$ then n can only be 0 or 1 and so on, then must be that

$$\frac{1}{1-w}e^{-tw/(1-w)} = \sum_{j=0}^{\infty}\sum_{n=0}^{j}\frac{(-1)^n}{n!}\binom{j}{n}w^j t^n$$

Or renaming variables $n \to j$ and $j \to n$ we get that

$$\frac{1}{1-w}e^{-tw/(1-w)} = \sum_{n=0}^{\infty}\sum_{j=0}^{n}\frac{(-1)^j}{j!}\binom{n}{j}t^j w^n$$

But from equation (10c) we know that

$$L_n(t) = \sum_{j=0}^{n}\frac{(-1)^j}{j!}\binom{n}{j}t^j$$

Therefore

$$\frac{1}{1-w}e^{-tw/(1-w)} = \sum_{n=0}^{\infty}L_n(t)w^n$$

$\square$

Proof. **12.**

(a) Let us differentiate ψ with respect to w

$$\begin{aligned}\frac{\mathrm{d}\psi(t,w)}{\mathrm{d}w} &= \frac{\mathrm{d}}{\mathrm{d}w}\left(\frac{1}{1-w}e^{-tw/(1-w)}\right)\\ &= \frac{1}{(1-w)^2}e^{-tw/(1-w)} - \frac{1}{(1-w)}\frac{te^{-tw/(1-w)}}{(1-w)^2}\\ &= \frac{1}{1-w}\sum_{n=0}^{\infty}L_n w^n - \frac{t}{(1-w)^2}\sum_{n=0}^{\infty}L_n w^n\end{aligned}$$

From the RHS we have that

$$\frac{\mathrm{d}}{\mathrm{d}w}\left(\sum_{n=0}^{\infty}L_n w^n\right) = \sum_{n=1}^{\infty}nL_n w^{n-1} = \sum_{n=0}^{\infty}(n+1)L_{n+1}w^n$$

So

$$(1-w)\sum_{n=0}^{\infty}L_n w^n - t\sum_{n=0}^{\infty}L_n w^n = (1-w)^2\sum_{n=0}^{\infty}(n+1)L_{n+1}w^n$$

Hence

$$\sum_{n=0}^{\infty}L_n w^n - \sum_{n=0}^{\infty}L_n w^{n+1} - t\sum_{n=0}^{\infty}L_n w^n =$$

$$= \sum_{n=0}^{\infty}(n+1)L_{n+1}w^n - 2\sum_{n=0}^{\infty}(n+1)L_{n+1}w^{n+1} + \sum_{n=0}^{\infty}(n+1)L_{n+1}w^{n+2}$$

$$(1-t)\sum_{n=0}^{\infty}L_n w^n - \sum_{n=1}^{\infty}L_{n-1}w^n =$$

$$= \sum_{n=0}^{\infty}(n+1)L_{n+1}w^n - 2\sum_{n=1}^{\infty}nL_n w^n + \sum_{n=2}^{\infty}(n-1)L_{n-1}w^n$$

If we make every summation to start from $n=2$ then

$$\begin{aligned}&A + (1-t)\sum_{n=2}^{\infty}L_n w^n - \sum_{n=2}^{\infty}L_{n-1}w^n =\\ &= B + \sum_{n=2}^{\infty}(n+1)L_{n+1}w^n - 2\sum_{n=2}^{\infty}nL_n w^n + \sum_{n=2}^{\infty}(n-1)L_{n-1}w^n\end{aligned}$$

Where

$$\begin{aligned}A &= (1-t)[L_0 + L_1 w] - L_0 w\\ B &= L_1 + 2L_2 w - 2L_1 w\end{aligned}$$

Considering only what happens for $n \geq 2$ we can collapse the summations into one summation which is equal to 0, then must be that each term is equal to 0, so

$$
\begin{aligned}
(n+1)L_{n+1} - 2nL_n + (n-1)L_{n-1} - (1-t)L_n + L_{n-1} &= 0 \\
(n+1)L_{n+1} - 2nL_n + nL_{n-1} - L_{n-1} - L_n + tL_n + L_{n-1} &= 0 \\
(n+1)L_{n+1} - (2n+1-t)L_n + nL_{n-1} &= 0
\end{aligned}
$$

The last equation is the relation we are looking for.

(b) Let us differentiate ψ with respect to t

$$
\begin{aligned}
\frac{\mathrm{d}\psi(t,w)}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{1-w}e^{-tw/(1-w)}\right) \\
&= -\frac{we^{-tw/(1-w)}}{(1-w)^2} \\
&= -\frac{w}{1-w}\sum_{n=0}^{\infty} L_n w^n
\end{aligned}
$$

From the RHS we have that

$$
\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_{n=0}^{\infty} L_n w^n\right) = \sum_{n=1}^{\infty} L_n' w^n
$$

Where we start from 1 now since $L_0 = 1$ then

$$
\begin{aligned}
-\frac{w}{1-w}\sum_{n=0}^{\infty} L_n w^n &= \sum_{n=1}^{\infty} L_n' w^n \\
-w\sum_{n=0}^{\infty} L_n w^n &= (1-w)\sum_{n=1}^{\infty} L_n' w^n \\
-\sum_{n=0}^{\infty} L_n w^{n+1} &= \sum_{n=1}^{\infty} L_n' w^n - \sum_{n=1}^{\infty} L_n' w^{n+1} \\
-\sum_{n=1}^{\infty} L_{n-1} w^n &= \sum_{n=1}^{\infty} L_n' w^n - \sum_{n=2}^{\infty} L_{n-1}' w^n \\
L_0 w + \sum_{n=2}^{\infty} L_{n-1} w^n &= -L_1' w + \sum_{n=2}^{\infty} L_{n-1}' w^n - \sum_{n=2}^{\infty} L_n' w^n
\end{aligned}
$$

If we only consider the case when $n \geq 2$ then we get the following relation

$$
L_{n-1} = L_{n-1}' - L_n'
$$

□

Proof. **13.**
Let us write equation (b) for $n+1$, we see that

$$\begin{aligned} L_n(t) &= L_n'(t) - L_{n+1}'(t) \\ L_{n+1}'(t) &= L_n'(t) - L_n(t) \end{aligned}$$

Also, taking the derivative of equation (a) with respect to t we get that

$$(n+1)L_{n+1}'(t) - (2n+1)L_n'(t) + tL_n'(t) + L_n(t) + nL_{n-1}'(t) = 0$$

Replacing $L_{n+1}'(t)$ we get that

$$\begin{aligned} (n+1)(L_n'(t) - L_n(t)) - (2n+1)L_n'(t) + tL_n'(t) + L_n(t) + nL_{n-1}'(t) &= 0 \\ nL_n'(t) + L_n'(t) - nL_n(t) - L_n(t) - 2nL_n'(t) - L_n'(t) + tL_n'(t) + L_n(t) + nL_{n-1}'(t) &= 0 \\ -nL_n(t) - nL_n'(t) + tL_n'(t) + nL_{n-1}'(t) &= 0 \end{aligned}$$

Now, applying again equation (b) to replace $L_{n-1}'(t)$ we have that

$$\begin{aligned} -nL_n(t) - nL_n'(t) + tL_n'(t) + n(L_{n-1}(t) + L_n'(t)) &= 0 \\ -nL_n(t) + tL_n'(t) + nL_{n-1}(t) &= 0 \end{aligned}$$

Therefore

$$tL_n'(t) = nL_n(t) - nL_{n-1}(t) \qquad (c)$$

We know that Laguerre's differential equation is

$$tL_n'' + (1-t)L_n' + nL_n = 0$$

Also, differentiating equation (c) with respect to t we have that

$$tL_n'' + L_n' = nL_n' - nL_{n-1}'$$

Replacing this and equation (c), we see that

$$\begin{aligned} tL_n'' + (1-t)L_n' + nL_n &= nL_n' - nL_{n-1}' - nL_n + nL_{n-1} + nL_n \\ &= nL_n' - nL_{n-1}' + nL_{n-1} \\ &= nL_n' - nL_{n-1}' + n(L_{n-1}' - L_n') \\ &= 0 \end{aligned}$$

Where we also replaced L_{n-1} with equation (b).
We see then that L_n satisfies Laguerre's differential equation. $\square$

Proof. **14.**
Let us consider the functions

$$e_n(t) = e^{-t/2} L_n(t)$$

We want to show they have norm 1 so we need to compute $\|e_n\| = \sqrt{\langle e_n, e_n \rangle}$ as follows

$$\|e_n\|^2 = \int_0^\infty e^{-t} L_n(t) L_n(t) \; dt$$

We know that $L_n(t)$ is a polynomial of degree n, also, we know from Problem 15 that if $m < n$ then $\int_0^\infty e^{-t} t^m L_n(t) \; dt = 0$ so the only term surviving from $L_n(t)$ in our integral is the term of degree n i.e.

$$\|e_n\|^2 = \int_0^\infty e^{-t} L_n(t) L_n(t) \; dt = \alpha \int_0^\infty e^{-t} t^n L_n(t) \; dt$$

Where α is the coefficient of t^n. We can determine it using equation (10c), where we see that $\alpha = (-1)^n/n!$. Then

$$\begin{aligned}
\|e_n\|^2 &= \frac{(-1)^n}{n!} \int_0^\infty e^{-t} t^n L_n(t) \; dt \\
&= \frac{(-1)^n}{n!} \int_0^\infty e^{-t} t^n \frac{e^t}{n!} f^{(n)}(t) \; dt \\
&= \frac{(-1)^n}{(n!)^2} \int_0^\infty t^n f^{(n)}(t) \; dt \\
&= \frac{(-1)^n}{(n!)^2} \left[t^n f^{(n-1)}(t) \Big|_0^\infty - n \int_0^\infty t^{n-1} f^{(n-1)}(t) \; dt \right] \\
&= \frac{n(-1)^{n+1}}{(n!)^2} \int_0^\infty t^{n-1} f^{(n-1)}(t) \; dt
\end{aligned}$$

Where we used that $f(t) = t^n e^{-t}$ and we integrated by parts. So, integrating by parts n times we get that

$$\|e_n\|^2 = \frac{n!(-1)^{2n}}{(n!)^2} \int_0^\infty f^{(0)}(t) \; dt = \frac{1}{n!} \int_0^\infty t^n e^{-t} \; dt$$

We apply integration by parts again setting $u = t^n$ and $dv = e^{-t} dt$, hence

$$\|e_n\|^2 = \frac{1}{n!} \left[-t^n e^{-t} \Big|_0^\infty + n \int_0^\infty t^{n-1} e^{-t} \; dt \right] = \frac{n}{n!} \int_0^\infty t^{n-1} e^{-t} \; dt$$

Therefore, again applying integration by parts n times we get that

$$\|e_n\|^2 = \frac{n!}{n!} \int_0^\infty e^{-t} \; dt = 1$$

$\square$

Proof. **15.**
We want to show that the functions

$$e_n(t) = e^{-t/2} L_n(t)$$

constitute an orthogonal sequence in the space $L^2[0, +\infty)$.
First, we want to compute the integral $\int_0^\infty e^{-t} t^m L_n(t)\ dt$ where $m < n$.
Let us define $f(t) = t^n e^{-t}$ so $L_n(t) = e^t f^{(n)}(t)/n!$, then

$$\begin{aligned}
\int_0^\infty e^{-t} t^m L_n(t)\ dt &= \frac{1}{n!} \int_0^\infty t^m f^{(n)}(t)\ dt \\
&= \frac{1}{n!} \left[t^m f^{(n-1)}(t) \Big|_0^\infty - \int_0^\infty m t^{m-1} f^{(n-1)}(t)\ dt \right] \\
&= -\frac{m}{n!} \int_0^\infty t^{m-1} f^{(n-1)}(t)\ dt
\end{aligned}$$

Where we applied integration by parts and we used that $f^{(n)}(t)$ is of the form $f^{(n)}(t) = e^{-t} p_n(t)$ where $p_n(t)$ is a polynomial of degree n and when $t \to \infty$ we see that $f^{(n)}(t) \to 0$.
Applying integration by parts m times we get that

$$\begin{aligned}
\int_0^\infty e^{-t} t^m L_n(t)\ dt &= (-1)^m \frac{m!}{n!} \int_0^\infty f^{(n-m)}(t)\ dt \\
&= (-1)^m (n-m)! \left[f^{(n-m-1)}(t) \right]_0^\infty
\end{aligned}$$

We note that since $n-m-1 < n$ then the $n-m-1$ derivative of $f(t) = t^n e^{-t}$ is not going to have yet a constant term so we have that $f^{(n-m-1)}(t)|_{t=0} = 0$ and we know that $f^{(n-m-1)}(t) \to 0$ when $t \to \infty$, therefore

$$\int_0^\infty e^{-t} t^m L_n(t)\ dt = 0$$

If we take $L_m(t)$ with $m < n$ then $L_m(t)$ is a polynomial of degree smaller than n and hence

$$\langle e_n(t), e_m(t) \rangle = \int_0^\infty e^{-t} L_m(t) L_n(t)\ dt = 0$$

Therefore the functions $e_n(t)$ constitute an orthogonal sequence in the space $L^2[0, +\infty)$. $\square$

3.8 - Functionals on Hilbert Spaces

Proof. **1.**
Let us consider the Hilbert space $\mathbb{R}^3$ with the dot product as our definition of inner product.
Let f be a linear functional on $\mathbb{R}^3$, if $x \in \mathbb{R}^3$ we see that

$$\begin{aligned}
|f(x)| &= |\xi_1 f(e_1) + \xi_2 f(e_2) + \xi_3 f(e_3)| \\
&= \left| \sum_{i=1}^{3} \xi_i f(e_i) \right| \\
&\le \sum_{i=1}^{3} |\xi_i f(e_i)| \\
&\le \sqrt{\sum_{i=1}^{3} |\xi_i|^2} \sqrt{\sum_{i=1}^{3} |f(e_i)|^2} \\
&\le \|x\| \sqrt{\sum_{i=1}^{3} |f(e_i)|^2}
\end{aligned}$$

Where we used the linearity of f and the Cauchy-Schwartz inequality.
So, there exists a real number $c = \sqrt{\sum_{i=1}^{3} |f(e_i)|^2}$ such that

$$|f(x)| \le c\|x\|$$

i.e. f is bounded.
Therefore, by Riesz's theorem we know that f can be represented in terms of the inner product

$$f(x) = \langle x, z \rangle = x \cdot z = \xi_1 \zeta_1 + \xi_2 \zeta_2 + \xi_3 \zeta_3$$

□

Proof. **2.**
Let f be a bounded linear functional on l^2, since l^2 is a Hilbert space with the inner product defined as

$$\langle x, y\rangle = \sum_{j=1}^{\infty} \xi_j \overline{\eta}_j$$

Then by Riesz's theorem we get that f can be represented in terms of the inner product as follows

$$f(x) = \langle x, z\rangle = \sum_{j=1}^{\infty} \xi_j \overline{\zeta}_j$$

Where $z = (\zeta_j) \in l^2$. $\square$

Proof. **3.**
Let z be any fixed element of an inner product space X and let us define $f(x) = \langle x, z \rangle$. If $x_1, x_2 \in X$ and α, β are scalars we see that

$$\begin{aligned} f(\alpha x_1 + \beta x_2) &= \langle \alpha x_1 + \beta x_2, z \rangle \\ &= \alpha \langle x_1, z \rangle + \beta \langle x_2, z \rangle \\ &= \alpha f(x_1) + \beta f(x_2) \end{aligned}$$

So f is a linear functional, also from Schwartz inequality we have that

$$|f(x)| = |\langle x, z \rangle| \leq \|x\| \|z\|$$

since z is fixed then setting $c = \|z\|$ shows us that f is bounded.
We know that the norm of f is defined as

$$\|f\| = \sup_{\substack{x \in X \\ x \neq 0}} \frac{|f(x)|}{\|x\|}$$

From the previous equation we note that

$$\|f\| = \sup_{\substack{x \in X \\ x \neq 0}} \frac{|f(x)|}{\|x\|} \leq \|z\|$$

Also, we know that $|f(x)| \leq \|f\| \|x\|$ so if we take $x = z$ we have that

$$|f(z)| = |\langle z, z \rangle| = \|z\|^2 \leq \|f\| \|z\|$$

Hence $\|z\| \leq \|f\|$, but then must be that

$$\|f\| = \|z\|$$

Therefore, f is a bounded linear functional on X of norm $\|z\|$. $\square$

Proof. **4.**
Let us consider the mapping $h : X \to X'$ such that $h : z \to f = \langle \cdot, z \rangle$ is surjective.
We know that X is an inner product space, so to show that X is a Hilbert space we need to show that X is complete.
We show first that h is injective, let us consider $f_1, f_2 \in X'$ which are associated to $z_1, z_2 \in X$ and suppose that $f_1 = f_2$ i.e. $\langle x, z_1 \rangle = \langle x, z_2 \rangle$ for all $x \in X$ then

$$\begin{aligned} \langle x, z_1 \rangle - \langle x, z_2 \rangle &= 0 \\ \langle x, z_1 - z_2 \rangle &= 0 \end{aligned}$$

If we take $x = z_1 - z_2$ then we get that

$$\langle x, z_1 - z_2 \rangle = \|z_1 - z_2\|^2 = 0$$

hence $z_1 - z_2 = 0$ so $z_1 = z_2$, which implies that h is injective and also bijective.
On the other hand, we know that the norm of f associated with z is $\|f\| = \| \langle \cdot, z \rangle \| = \|z\|$ so the norm is preserved.
Let $(f_{z_n}) \subset X'$ be a Cauchy sequence, where each functional f_{z_n} is associated to a $z_n \in X$, then $\|f_{z_n} - f_{z_m}\| \to 0$. Since each functional is linear then we have that

$$\|f_{z_n} - f_{z_m}\| = \|f_{z_n - z_m}\| = \|z_n - z_m\|$$

This shows us that (z_n) is also a Cauchy sequence in X.
But since the space X' (the dual space) is complete then the sequence (f_{z_n}) must converge, which implies that (z_n) converges.
Therefore X must be a Hilbert space. $\square$

Proof. **5.**
Let us consider the real space l^2.
By definition the dual space of l^2, named $(l^2)'$, is the set of all bounded linear functionals on l^2.
So if $f \in (l^2)'$ then by Riesz's theorem we know that f can be represented as $f(x) = \langle x, z \rangle$.
We want to show that the map $h : z \to \langle \cdot, z \rangle$ is an isomorphism.
We note that for every element $\langle \cdot, z \rangle \in (l^2)'$ there is an associated $z \in l^2$, so the map h is surjective.
We proved in Problem 4 that h is injective, and therefore it's bijective.
Also, the norm is preserved since by Riesz's theorem we know that $\|f\| = \|z\|$.
Therefore, h is an isomorphism between l^2 and $(l^2)'$ which implies that the dual space of l^2 is l^2. $\square$

Proof. **6.**
We have shown in Problem 4 and 5 that the map $z \to f_z = \langle \cdot, z \rangle$ is a bijection. Also, by Riesz's Theorem we know that $\|z\| = \|f_z\|$ so it's actually an isometric bijection.
We note that if $\alpha z + \beta v \in H$ then the map takes this element to $f_{\alpha z + \beta v} = \langle \cdot, \alpha z + \beta v \rangle \in H'$ and hence

$$f_{\alpha z+\beta v} = \langle \cdot, \alpha z + \beta v \rangle = \langle \cdot, \alpha z \rangle + \langle \cdot, \beta v \rangle = \overline{\alpha} \langle \cdot, z \rangle + \overline{\beta} \langle \cdot, v \rangle = \overline{\alpha} f_z + \overline{\beta} f_v$$

Where we used the properties of the inner product.
Therefore, the map is conjugate linear.

$$|a\rangle \to \langle \cdot | a \rangle$$

□

Proof. **7.**
We know that the dual space of a normed space is a Banach space, so since H is a Hilbert space and hence has an inner product that defines a norm on it then the dual space H' is a Banach space i.e. it's complete.
In addition to that, let us define the inner product map on H' as

$$\langle f_z, f_v \rangle_1 = \overline{\langle z, v \rangle} = \langle v, z \rangle$$

where $f_z(x) = \langle x, z \rangle$ and $f_v(x) = \langle x, v \rangle$.
Since, essentially the inner product $\langle f_z, f_v \rangle_1$ is defined in terms of the inner product of the elements $v, z \in H$ then it must satisfy every property of the inner product and hence it's a proper inner product over H'.
Therefore, H' is a Hilbert space. □

Proof. **8.**
From Problem 6 we know that there exists an isometric bijection $z \to f_z$ from H to H', also, from Problem 7 we know that the dual space H' is also a Hilbert space with the inner product defined as

$$\langle f_z, f_v \rangle_1 = \langle v, z \rangle$$

If we consider now the second dual space $H'' = (H')'$ because of Riesz's theorem we know that any bounded linear functional on H' can be represented in terms of the inner product of H', namely

$$g_{f_z}(f_x) = \langle f_x, f_z \rangle_1 = \langle z, x \rangle$$

Where the norm has the following property $\|g_{f_z}\| = \|f_z\| = \|z\|$.
Let us consider now the map $h : z \to g_{f_z}$. Suppose $g_{f_v} = g_{f_z}$ then

$$\begin{aligned} \langle f_x, f_v \rangle - \langle f_x, f_z \rangle &= 0 \\ \langle v, x \rangle - \langle z, x \rangle &= 0 \\ \langle v - z, x \rangle &= 0 \end{aligned}$$

If we let $x = v - z$ then

$$\langle v - z, v - z \rangle = \|v - z\|^2 = 0$$

Then $v - z = 0$ so $v = z$, which implies that the map h is injective.
Also, we know that the map $z \to f_z$ is surjective, then the map $f_z \to g_{f_z}$ is also surjective and hence h is surjective and therefore bijective.
Finally, if we define the norm on H'' as we did for H' we get that

$$\langle g_{f_v}, g_{f_z} \rangle_2 = \langle f_z, f_v \rangle_1 = \langle v, z \rangle$$

Which implies that the map h is an isomorphism between H and H'' and therefore H and H'' are isomorphic. $\square$

Proof. **10.**
Let us consider the inner product $\langle \cdot, \cdot \rangle$ as a map from $X \times X$ to a field K where X is any inner product space.
Let $x, y, x_1, y_1, x_2, y_2 \in X$ then, we note that

$$\begin{aligned}
\langle x_1 + x_2, y \rangle &= \langle x_1, y \rangle + \langle x_2, y \rangle \\
\langle x, y_1 + y_2 \rangle &= \langle x, y_1 \rangle + \langle x, y_2 \rangle \\
\langle \alpha x, y \rangle &= \alpha \langle x, y \rangle \\
\langle x, \beta y \rangle &= \overline{\beta} \langle x, y \rangle
\end{aligned}$$

Where we used only the inner product properties.
By Schwarz inequality we have that

$$| \langle x, y \rangle | \leq \|x\| \|y\|$$

Therefore $\langle \cdot, \cdot \rangle$ is a bounded sesquilinear form.
We see that

$$\| \langle \cdot, \cdot \rangle \| = \sup_{\substack{x \in X \\ x \neq 0}} \frac{| \langle x, y \rangle |}{\|x\| \|y\|} \leq 1$$

Also, we know that

$$| \langle x, y \rangle | \leq \| \langle \cdot, \cdot \rangle \| \|x\| \|y\|$$

Then taking $y = x$ and dividing by $\|x\| \|y\|$ we get that

$$1 = \frac{\|x\|^2}{\|x\| \|x\|} = \frac{| \langle x, x \rangle |}{\|x\| \|x\|} \leq \| \langle \cdot, \cdot \rangle \|$$

Therefore, must be that

$$\| \langle \cdot, \cdot \rangle \| = 1$$

$\square$

Proof. **11.**
Let X be a vector sapce and h a sesquilinear form on $X \times X$. We want to show that $f_1(x) = h(x, y_0)$ with fixed y_0 defines a linear functional f_1 on X and so does $f_2(y) = \overline{h(x_0, y)}$ with fixed x_0.
Since by definition $h : X \times X \to K$ then given a fixed $y_0 \in X$ we see that $f_1 : X \to K$.
Let $x_1, x_2 \in X$ and $\alpha, \beta \in K$ where K is a field, then we see that

$$\begin{aligned} f_1(\alpha x_1 + \beta x_2) &= h(\alpha x_1 + \beta x_2, y_0) \\ &= \alpha h(x_1, y_0) + \beta h(x_2, y_0) \\ &= \alpha f_1(x_1) + \beta f_1(x_2) \end{aligned}$$

Where we used the properties of the sesquilinear form h.
Therefore f_1 defines a linear functional on X.
In the case of f_2, given $y_1, y_2 \in X$, we see that

$$\begin{aligned} f_2(\alpha y_1 + \beta y_2) &= \overline{h(x_0, \alpha y_1 + \beta y_2)} \\ &= \overline{\overline{\alpha} h(x_0, y_1)} + \overline{\overline{\beta} h(x_0, y_2)} \\ &= \alpha \overline{h(x_0, y_1)} + \beta \overline{h(x_0, y_2)} \\ &= \alpha f_2(y_1) + \beta f_2(y_2) \end{aligned}$$

Therefore f_2 also defines a linear functional on X. $\square$

Proof. **12.**
Let X and Y be normed spaces and let $h : X \times Y \to K$ be a bounded sesquilinear form.
Let us consider $x_0 \in X$, $y_0 \in Y$ and $\epsilon > 0$ then let us consider all the $x \in X$ and $y \in Y$ such that

$$\|x - x_0\| < \delta \quad \text{and} \quad \|y - y_0\| < \delta$$

Where

$$\delta = \frac{\epsilon}{\|h\|(1 + \|x_0\| + \|y_0\|)}$$

We note that

$$\begin{aligned} |h(x,y) - h(x_0,y_0)| &= |h(x,y) - h(x_0,y) + h(x_0,y) - h(x_0,y_0)| \\ &= |h(x - x_0, y) + h(x_0, y - y_0)| \\ &\leq |h(x - x_0, y)| + |h(x_0, y - y_0)| \end{aligned}$$

Since h is bounded we have that

$$\begin{aligned} |h(x,y) - h(x_0,y_0)| &\leq |h(x - x_0, y)| + |h(x_0, y - y_0)| \\ &\leq \|h\|\|x - x_0\|\|y\| + \|h\|\|x_0\|\|y - y_0\| \\ &\leq \|h\|\|x - x_0\|\|y - y_0\| + \|h\|\|x - x_0\|\|y_0\| + \|h\|\|x_0\|\|y - y_0\| \\ &< \|h\|(\delta^2 + \delta(\|y_0\| + \|x_0\|)) \end{aligned}$$

Where we used that

$$\|y\| \leq \|y - y_0\| + \|y_0\|$$

If we choose $\delta < 1$ then $\delta^2 < \delta$ and hence

$$|h(x,y) - h(x_0,y_0)| < \|h\|\delta(1 + \|y_0\| + \|x_0\|) = \epsilon$$

Therefore h is jointly continuous. $\square$

Proof. **13.**
Let $h : X \times X \to K$ be a Hermitian sesquilinear form, then if $K = \mathbb{R}$ we have that $\overline{h(y,x)} = h(y,x)$ and hence the last condition is

$$h(x,y) = h(y,x)$$

We note that

$$\begin{aligned} h(x,\alpha y) &= h(\alpha y, x) = \alpha h(y,x) = \alpha h(x,y) \\ h(z, x+y) &= h(x+y, z) = h(x,z) + h(y,z) = h(z,x) + h(z,y) \end{aligned}$$

So h becomes a bilinear function.
For h to be an inner product we need to add the following conditions

$$\begin{aligned} &h(x,x) \geq 0 \\ &h(x,x) = 0 \quad \text{if and only if} \quad x = 0 \end{aligned}$$

□

Proof. **14.**
Let X be a vector space and h a Hermitian form on $X \times X$ such that it is positive semidefinite i.e. $h(x,x) \geq 0$ for all $x \in X$.
Let $x, y \in X$ and $\alpha \in K$, and let us consider the case when $h(y,y) > 0$.
We note that

$$\begin{aligned}
0 \leq h(x-\alpha y, x-\alpha y) &= h(x, x-\alpha y) - \alpha h(y, x-\alpha y) \\
&= \overline{h(x-\alpha y, x)} - \alpha \overline{h(x-\alpha y, y)} \\
&= \overline{h(x,x)} - \overline{\alpha h(y,x)} - \alpha \overline{h(x,y)} + \alpha \overline{\alpha h(y,y)} \\
&= h(x,x) - \overline{\alpha} h(x,y) - \alpha h(y,x) + \alpha \overline{\alpha} h(y,y) \\
&= h(x,x) - \overline{\alpha} h(x,y) - \alpha [h(y,x) - \overline{\alpha} h(y,y)]
\end{aligned}$$

Let us choose $\overline{\alpha} = h(y,x)/h(y,y)$, then we get that

$$\begin{aligned}
0 &\leq h(x,x) - \frac{h(y,x)}{h(y,y)} h(x,y) \\
0 &\leq h(x,x) - \frac{\overline{h(x,y)}}{h(y,y)} h(x,y) \\
0 &\leq h(x,x) - \frac{|h(x,y)|^2}{h(y,y)}
\end{aligned}$$

Multiplying by $h(y,y)$ we have that

$$0 \leq h(x,x)h(y,y) - |h(x,y)|^2$$

Hence

$$|h(x,y)|^2 \leq h(x,x)h(y,y)$$

If $h(y,y) = 0$ we may repeat the same argument but considering instead $h(y - \alpha x, y - \alpha x)$, for which we get that

$$0 \leq h(y-\alpha x, y-\alpha x) = -\overline{\alpha} h(y,x) - \alpha [h(x,y) - \overline{\alpha} h(x,x)]$$

And taking $\overline{\alpha} = h(x,y)/h(x,x)$ we get that

$$\begin{aligned}
0 &\leq -\frac{h(x,y)}{h(x,x)} h(y,x) \\
|h(x,y)|^2 &\leq 0
\end{aligned}$$

Therefore $|h(x,y)|^2 = 0$ which is the expected result.
Finally, the last case to take into account is $h(x,x) = h(y,y) = 0$. We see that

$$\begin{aligned}
0 \leq h(x+y, x+y) &= h(x,x) + h(x,y) + h(y,x) + h(y,y) \\
&= h(x,y) + h(y,x)
\end{aligned}$$

and that

$$\begin{aligned}0 \leq h(x-y, x-y) &= h(x,x) - h(x,y) - h(y,x) + h(y,y) \\ &= -h(x,y) - h(y,x)\end{aligned}$$

Then $h(x,y) + h(y,x) \leq 0$, therefore, must be that

$$\begin{aligned}h(x,y) + h(y,x) &= 0 \\ h(x,y) + \overline{h(x,y)} &= 0\end{aligned}$$

This implies that $\mathrm{Re}\{h(x,y)\} = 0$.
Also, let us consider $h(ix, y)$, we note that

$$\begin{aligned}0 &= \mathrm{Re}\{h(ix,y)\} \\ &= \mathrm{Re}\{ih(x,y)\} \\ &= \mathrm{Re}\{i(\mathrm{Re}\{h(x,y)\} + i\,\mathrm{Im}\{h(x,y)\})\} \\ &= \mathrm{Re}\{i\,\mathrm{Re}\{h(x,y)\} - \mathrm{Im}\{h(x,y)\}\} \\ &= -\,\mathrm{Im}\{h(x,y)\}\end{aligned}$$

Therefore $h(x,y) = 0$ and hence the Schwarz inequality

$$|h(x,y)|^2 \leq h(x,x)h(y,y)$$

is satisfied. $\square$

Proof. **15.**
Let h be a Hermitian form such that $h(x,x) \geq 0$ for every $x \in X$ and let us define $p(x) = +\sqrt{h(x,x)}$.
We see that $p(x) \geq 0$ by definition so $p(x)$ satisfies (N1) property.
We note that

$$p(\alpha x) = \sqrt{h(\alpha x, \alpha x)} = \sqrt{\alpha\overline{\alpha}h(x,x)} = \sqrt{|\alpha|^2 h(x,x)} = |\alpha|\sqrt{h(x,x)} = |\alpha|p(x)$$

So $p(x)$ satisfies (N3) property. Also, let $x, y \in X$, we see that

$$p(x+y)^2 = h(x+y, x+y) = h(x,x) + 2\operatorname{Re}\{h(x,y)\} + h(y,y)$$

Since $\operatorname{Re}\{h(x,y)\} \leq |h(x,y)|$ then

$$p(x+y)^2 \leq h(x,x) + 2|h(x,y)| + h(y,y)$$

But also because Schwarz inequality is satisfied by h we have that

$$\begin{aligned} p(x+y)^2 &\leq h(x,x) + 2|h(x,y)| + h(y,y) \\ &\leq h(x,x) + 2\sqrt{h(x,x)h(y,y)} + h(y,y) \\ &\leq p(x)^2 + 2p(x)p(y) + p(y)^2 \\ &\leq (p(x) + p(y))^2 \end{aligned}$$

Therefore

$$p(x+y) \leq p(x) + p(y)$$

So $p(x)$ satisfies (N4) property and hence $p(x)$ defines a seminorm on X. □